# UNIT I : Introduction to Heat Transfer

Subject: Heat Transfer (23A03601T) Topics: Modes, conduction eqn, slabs, cylinders, critical radius, fins, transient conduction

## 1.1 What is Heat Transfer?

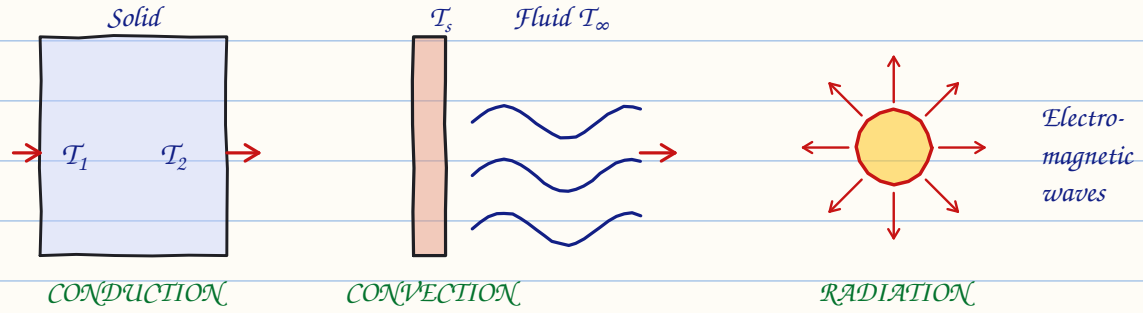
Heat transfer is the study of energy in transit because of a temperature difference. Thermodynamics tells us how much energy is transferred (end states, equilibrium); heat transfer tells us how fast (rate) and by what mechanism.

→ Driving potential : temperature difference $\Delta T$ (like voltage in electricity)

→ Heat always flows from a high temperature region to a low temperature region (2nd law)

→ Unit of heat rate $Q$ : Watt ($W = J/s$); heat flux $q = Q/A$ : $W/m^2$

## 1.2 Basic Modes of Heat Transfer




| Mode | Rate equation | Medium | Key property |
|---|---|---|---|
| Conduction | $Q = -kA\, dT/dx$ | Solid / fluid at rest | $k$ (W/mK) |
| Convection | $Q = hA(T_s - T_\infty)$ | Moving fluid | $h$ ($W/m^2K$) |
| Radiation | $Q = \varepsilon\sigma A(T_1^4 - T_2^4)$ | No medium (vacuum) | $\varepsilon$, $\sigma = 5.67\times10^{-8}$ |

## 1.3 Rate Equations

(a) Conduction – Fourier's law: rate of heat conduction is proportional to the area normal to flow and the temperature gradient.

$$Q = -kA\,(dT/dx)\ [W] \qquad \text{Fourier law}$$

→ $k$ = thermal conductivity; -ve sign : heat flows in direction of falling $T$

→ $k$: metals (Cu 385, Al 205, Steel 45) > liquids (water 0.6) > gases (air 0.026) ; insulators 0.03–0.1

(b) Convection – Newton's law of cooling:

$$Q = h\,A\,(T_s - T_\infty)\ [W]$$ Newton law

→ $h$ = convective heat transfer coefficient ($W/m^2K$); depends on velocity, fluid properties, geometry

→ Free convection $h \approx 2$–$25$ ; forced (air) 25–250 ; forced (water) 50–20,000 ; boiling/condensation 2500–100,000

(c) Radiation – Stefan–Boltzmann law:

$$E_b = \sigma\,T^4\ [W/m^2]$$

$$Q = \varepsilon\,\sigma\,A\,(T_1^4 - T_2^4)$$ Stefan–Boltzmann

→ $\sigma = 5.67\times10^{-8}\ W/m^2K^4$; $T$ in Kelvin; $\varepsilon$ = emissivity (0 to 1)

## 1.4 Thermal Diffusivity

$$\alpha = k / (\rho\,c_p)\ [m^2/s]$$

Ratio of heat conducted to heat stored. High $\alpha$ → heat diffuses quickly through the material (e.g., metals); low $\alpha$ → slow response.

Note: In real problems all three modes occur together (combined mode). e.g. heat loss from a hot pipe: conduction in the wall, convection to air and radiation to surroundings.

## Thermal Conductivity – a closer look

$k$ is a property : the heat conducted per unit area per unit time for unit temperature gradient. Mechanism: free-electron flow + lattice vibrations (metals), lattice vibrations (insulating solids), molecular collisions (fluids).

| Material | $k$ at 20°C (W/mK) | Material | $k$ at 20°C (W/mK) |
|---|---|---|---|
| Copper | 385 | Water | 0.60 |
| Aluminium | 205 | Engine oil | 0.145 |
| Mild steel | 45 | Air | 0.026 |
| Brick (common) | 0.72 | Glass wool | 0.04 |
| Concrete | 1.4 | Asbestos sheet | 0.16 |

→ $k$ varies with temperature : $k = k_0(1 + \beta T)$. For pure metals $k$ decreases with $T$; for gases $k$ increases with $T$; for most insulators $k$ increases with $T$

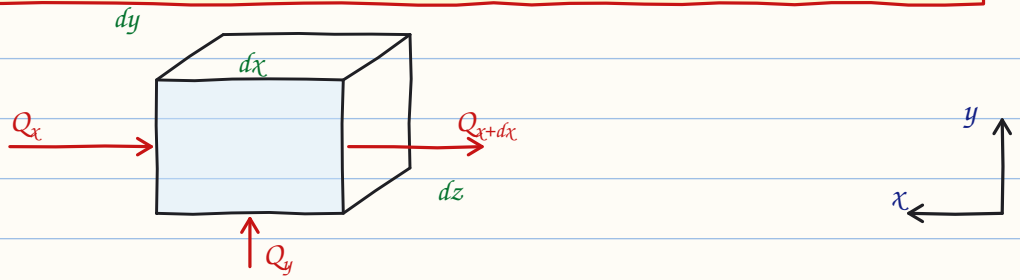
→ Impurities and alloying reduce $k$ of metals (pure Cu 385 → brass 110); moisture in insulation increases $k$

→ Mean $k$ is used if variation is small: $Q = k_m\,A\,\Delta T/L$ with $k_m = k_0[1 + \beta(T_1+T_2)/2]$

## 1.5 Generalised Heat Conduction Equation (Cartesian)

Consider an elemental volume $dx \cdot dy \cdot dz$ in a homogeneous solid with internal heat generation $q_g$ ($W/m^3$). Energy balance on the element:

**(Net heat conducted in) + (Heat generated) = (Rate of increase of internal energy)**




Heat in along $x$ : $Q_x = -k\,dy\,dz\,(\partial T/\partial x)$. Heat out : $Q_{x+dx} = Q_x + \partial/\partial x\,(Q_x)\,dx$. Net along $x = \partial/\partial x\,(k\,\partial T/\partial x)\,dx\,dy\,dz$. Similarly for y and z. Generation = $q_g\,dx\,dy\,dz$. Storage = $\rho\,c_p\,(\partial T/\partial t)\,dx\,dy\,dz$. Hence,

$$\partial/\partial x(k\,\partial T/\partial x) + \partial/\partial y(k\,\partial T/\partial y) + \partial/\partial z(k\,\partial T/\partial z) + q_g = \rho\,c_p\,\partial T/\partial t$$

## 1.6 Various Forms of the Conduction Equation

Constant $k$ (homogeneous, isotropic): with $\alpha = k/\rho c_p$

$$\nabla^2 T + q_g/k = (1/\alpha)\,\partial T/\partial t$$ *Fourier–Biot eqn*

| Condition | Equation | Name |
|---|---|---|
| No generation, unsteady | $\nabla^2 T = (1/\alpha)\,\partial T/\partial t$ | Fourier eqn |
| Steady, with generation | $\nabla^2 T + q_g/k = 0$ | Poisson eqn |
| Steady, no generation | $\nabla^2 T = 0$ | Laplace eqn |
| 1-D steady, no generation | $d^2T/dx^2 = 0$ | Linear T profile |

Cylindrical co-ordinates ($r$, $\varphi$, $z$):

$$(1/r)\,\partial/\partial r(\,r\,\partial T/\partial r\,) + (1/r^2)\,\partial^2 T/\partial \varphi^2 + \partial^2 T/\partial z^2 + q_g/k = (1/\alpha)\,\partial T/\partial t$$

Spherical co-ordinates ($r$, $\theta$, $\varphi$):

$$(1/r^2)\,\partial/\partial r(\,r^2\,\partial T/\partial r\,) + (1/r^2\sin^2\theta)\,\partial^2 T/\partial \varphi^2 + (1/r^2\sin\theta)\,\partial/\partial\theta(\,\sin\theta\,\partial T/\partial\theta\,) + q_g/k = (1/\alpha)\,\partial T/\partial t$$

1-D steady radial forms (no generation): Cylinder: $d/dr(\,r\,dT/dr\,) = 0$ Sphere: $d/dr(\,r^2\,dT/dr\,) = 0$

## 1.7 Initial and Boundary Conditions

→ Initial condition : $T(x, 0) = T_i$ (needed for unsteady problems)

→ 1st kind (Dirichlet) : surface temperature specified, $T(0,t) = T_s$

→ 2nd kind (Neumann) : heat flux specified, $-k\,\partial T/\partial x\,|_0 = q_0$; insulated surface : $\partial T/\partial x = 0$

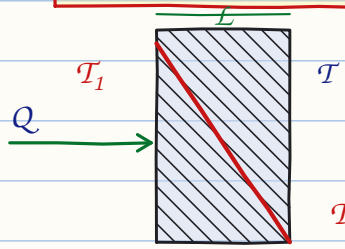
→ 3rd kind (Robin / convection) : $-k\,\partial T/\partial x\,|_0 = h\,[\,T_\infty - T(0)\,]$

# Steady Conduction – Plane Wall

Wall thickness $L$, area $A$, constant $k$, no generation, faces at $T_1$ and $T_2$. From $d^2T/dx^2 = 0$ : $T = C_1x + C_2$. Using $T(0)=T_1$, $T(L)=T_2$:

$$T(x) = T_1 - (T_1 - T_2)\,x / L$$

$$Q = k A\,(T_1 - T_2) / L = (T_1 - T_2) / (L / kA)$$ *linear profile*




T-profile is a straight line

Analogy: $I = \Delta V / R$

$Q = \Delta T / R_{th}$

$R_{th} = L / kA$




→ Thermal resistance : conduction $R = L/kA$ (K/W); convection $R = 1/hA$; radiation $R = 1/h_rA$

## Composite (series) wall

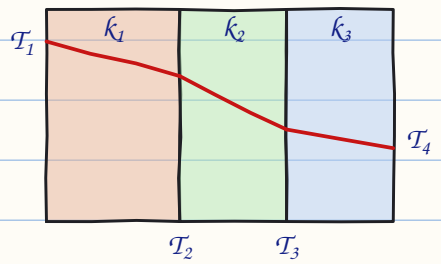
Layers 1,2,3 in series carry the same Q. Add resistances like electrical resistors:

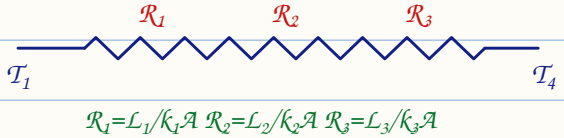
$$Q = (T_1 - T_4) / (\,L_1/k_1A + L_2/k_2A + L_3/k_3A\,)$$

With convection on both sides (hot fluid $T_h$, cold fluid $T_c$) :

$$Q = (T_h - T_c) / [\,1/h_1A + \Sigma L/kA + 1/h_2A\,] = U\,A\,(T_h - T_c)$$

$$U = 1 / (\,1/h_1 + \Sigma L/k + 1/h_2\,)\ [W/m^2K]$$ *overall U*




$R_1 = L_1/k_1A$ $R_2 = L_2/k_2A$ $R_3 = L_3/k_3A$

Interface temperatures : $T_2 = T_1 - Q R_1$, $T_3 = T_2 - Q R_2$. Parallel layers : $1/R_{eq} = 1/R_a +$

$1/R_t$ (same $\Delta T$, $Q$ adds).

> **Note:** Contact resistance : imperfect contact between two solids creates a temperature drop $\Delta T_c$ at the interface, $R_c = \Delta T_c/Q$ ($m^2K/W$). It is reduced by increasing pressure or using thermal grease / soft foil.

## Solved Example 1 – Furnace wall

Wall: fire-brick $L$=0.25 m, $k$=1.0; insulation $L$=0.10 m, $k$=0.05; red brick $L$=0.20 m, $k$=0.8 W/mK. Inner face 1000°C, outer face 50°C. Find heat loss per $m^2$ and interface temperatures.

$R/A = 0.25/1 + 0.10/0.05 + 0.20/0.8 = 0.25 + 2.0 + 0.25 =$ **2.5 $m^2K/W$**

$q = (1000-50)/2.5 =$ **380 $W/m^2$**; $T_2 = 1000 - 380(0.25) =$ **905°C**; $T_3 = 905 - 380(2.0) =$ **145°C**

## Plane wall with internal heat generation

Slab of thickness $2L$, both faces held at $T_w$, uniform generation $q_g$ ($W/m^3$). From $d^2T/dx^2 + q_g/k = 0$ with symmetry ($dT/dx = 0$ at $x = 0$):

$$T(x) = T_w + (q_g/2k)(L^2 - x^2)$$

$$T_{max} = T_w + q_gL^2/2k \text{ (at the mid-plane)} \qquad \text{parabolic}$$

→ Long solid cylinder (radius $R$): $T = T_w + (q_gR^2/4k)(1 - r^2/R^2)$, $T_{max} = T_w + q_gR^2/4k$

→ Solid sphere: $T_{max} = T_w + q_gR^2/6k$. Heat generated per unit volume = heat lost at the surface (steady).

## Solved Example 1B – Wall with convection on both sides

Brick wall $L$ = 0.3 m, $k$ = 0.6 W/mK. Inside air 25°C ($h_i$ = 8), outside air -5°C ($h_o$ = 20 $W/m^2K$). Find $U$, heat loss and inside surface temperature.

$R = 1/8 + 0.3/0.6 + 1/20 = 0.125 + 0.5 + 0.05 = 0.675\ m^2K/W \rightarrow$ **$U = 1.48\ W/m^2K$**

$q = 30/0.675 =$ **44.4 $W/m^2$**; $T_{s,in} = 25 - 44.4(0.125) =$ **19.4°C**

# Steady Conduction – Cylinders & Spheres

Hollow cylinder: inner radius $r_1$ ($T_1$), outer radius $r_2$ ($T_2$), length $L$. Radial conduction, no generation: $d/dr(\ r\ dT/dr\ ) = 0 \rightarrow T = C_1 \ln r + C_2$. Heat flow through any radius is constant, $Q = -k(2\pi rL)\ dT/dr$. Integrating from $r_1$ to $r_2$:

$$Q = 2\pi k L (T_1 - T_2) / \ln(r_2/r_1)$$

$$R_{cyl} = \ln(r_2/r_1) / (2\pi k L)$$ cylinder

→ Temperature profile is logarithmic (not linear): $(T - T_1)/(T_2 - T_1) = \ln(r/r_1) / \ln(r_2/r_1)$

→ Heat flux $q = Q/A$ decreases with r (area $2\pi rL$ increases) but Q stays constant

Composite cylinder (pipe + insulation) with convection inside ($h_i$) and outside ($h_o$):

$$Q = (T_i - T_o) / [\ 1/(h_i A_i) + \ln(r_2/r_1)/(2\pi k_1 L) + \ln(r_3/r_2)/(2\pi k_2 L) + 1/(h_o A_o)\ ]$$

Hollow sphere: $d/dr(\ r^2\ dT/dr\ ) = 0 \rightarrow T$ varies as $1/r$:

$$Q = 4\pi k (T_1 - T_2) / (\ 1/r_1 - 1/r_2\ )$$

$$R_{sph} = (r_2 - r_1) / (4\pi k r_1 r_2)$$ sphere

## Solved Example 2 – Insulated steam pipe

Steam pipe, outer radius $r_1$=50 mm at 200°C, covered with 50 mm insulation (k=0.1 W/mK); outer surface of insulation at 40°C. Heat loss per metre?

$r_2$ = 100 mm. $Q/L = 2\pi k (T_1-T_2) / \ln(r_2/r_1) = 2\pi(0.1)(160) / \ln 2 = 100.5/0.693 =$ 145 W/m

## Solved Example 2B – Pipe with insulation and convection

Pipe surface 250°C, $r_1$ = 30 mm, covered with 25 mm of insulation (k = 0.08 W/mK). Ambient air 30°C, $h_o$ = 12 W/m²K. Find the loss per metre and the outer surface temperature of the insulation.

$r_2$ = 55 mm; $R_{ins} = \ln(55/30) / (2\pi \times 0.08) = 0.606/0.503 = 1.206$ mK/W

$R_{conv} = 1/(2\pi\ r_2\ h_o) = 1/(2\pi \times 0.055 \times 12) = 0.241$ mK/W

$Q/L = (250 - 30)/(1.206 + 0.241) =$ 152 W/m; $T_{surf} = 30 + 152(0.241) =$ 66.7°C

# Critical Thickness (Radius) of Insulation

Adding insulation to a small pipe increases the conduction resistance but also increases the outer surface area (lower convection resistance). So heat loss may first rise and then fall.

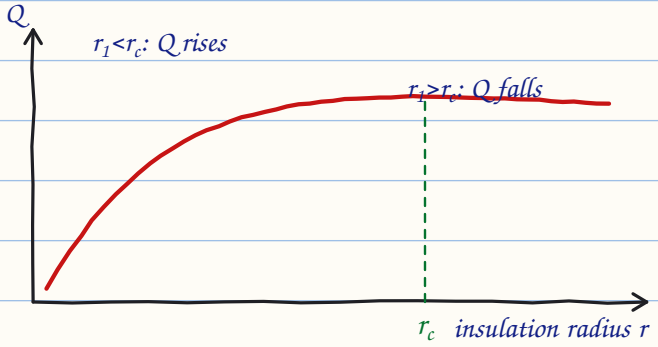
Pipe radius $r_1$, insulation up to radius r, conductivity k, outside convection $h_o$:

$$R_{tot} = \ln(r/r_1) / (2\pi k L) + 1 / (2\pi\ r\ L\ h_o)$$

For maximum heat loss (minimum $R_{tot}$) put $dR_{tot}/dr = 0$ : $1/(2\pi kLr) - 1/(2\pi\ r^2\ L\ h_o) = 0$

$$r_c = k / h_o \text{ (cylinder)}$$
$$r_c = 2k / h_o \text{ (sphere)}$$
critical radius

$d^2R/dr^2 > 0$ at $r = r_c$, so $R$ is a *minimum* and $Q$ a *maximum* there.




$r_1 < r_c$ : insulation *raises* loss

$r_1 > r_c$ : insulation *reduces* loss

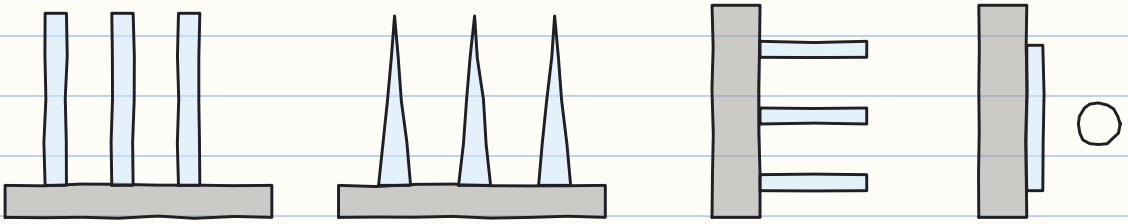
*Insulation effective only if* $r_1 \geq r_c$

→ Critical thickness = $r_c - r_1$. Used in electric wires (insulation cools them, $r_1<r_c$) and small tubes.

*Note:* For steam pipes $r_1$ is large ($r_1 > k/h \approx 20–50$ mm) so any insulation reduces the loss. Example: $k=0.5$, $h=10 \rightarrow r_c = 0.05$ m = 50 mm.

# Heat Conduction through Fins

A *fin* (extended surface) is a protrusion that increases the heat dissipation area: $Q = hA(T_s-T_\infty)$ rises when $A$ rises. Used on IC-engine cylinders, radiators, electronic heat sinks, condensers, transformers.




## Assumptions & governing equation

→ 1-D steady conduction along fin; constant $k$, $h$, $T_\infty$; no generation; negligible radiation; fin thin (Biot small)

Energy balance on element $dx$: $Q_x = Q_{x+dx} + h\,P\,dx\,(T - T_\infty)$. Put $\theta = T - T_\infty$, $P$ = perimeter, $A_c$ = cross-section area:

$$d^2\theta/dx^2 - m^2\,\theta = 0\,,\; m = \sqrt{(hP / kA_c)}$$
$$\theta = C_1\, e^{mx} + C_2\, e^{-mx}$$
fin equation

Base condition $\theta(0) = \theta_b = T_b - T_\infty$. Tip condition gives three cases (fin length $L$):

| Tip condition | $\theta/\theta_b$ | Heat dissipated Q |
|---|---|---|
| Very long fin | $e^{-mx}$ | $\sqrt{(hPkA_c)}\,\theta_b$ |
| Insulated tip | $\cosh m(L-x) / \cosh mL$ | $\sqrt{(hPkA_c)}\,\theta_b \tanh mL$ |
| Convecting tip | see formulas below | see formulas below |

$$\theta/\theta_b = [\cosh m(L-x) + (h/mk)\sinh m(L-x)] / [\cosh mL + (h/mk)\sinh mL]$$

$$Q = \sqrt{(hPkA_c)}\,\theta_b\,[\sinh mL + (h/mk)\cosh mL] / [\cosh mL + (h/mk)\sinh mL]$$

Note: Convecting tip is handled easily with corrected length in the insulated-tip formula : $L_c = L + t/2$ (rectangular fin, thickness t), $L_c = L + D/4$ (pin fin).

## Fin efficiency & effectiveness

$$\eta_{fin} = Q_{fin} / Q_{max} = Q_{fin} / (h A_{fin} \theta_b)$$ (insulated tip : $\eta = \tanh mL / mL$)

$$\varepsilon_{fin} = Q_{fin} / (h A_c \theta_b)$$ (very long fin : $\varepsilon = \sqrt{(kP / hA_c)}$) — $\eta$ and $\varepsilon$

→ $Q_{max}$ = heat if the whole fin were at base temperature. $\eta \to 1$ as $mL \to 0$ ; $\eta$ drops for long fins

→ $\varepsilon$ is the gain from adding the fin: $\varepsilon = \eta\,(A_{fin}/A_c)$. Fin is worthwhile only if $\varepsilon > 2$

→ To raise $\varepsilon$: high k material (Cu, Al), thin closely spaced fins, fins on the side with low h (gas side), large $P/A_c$

## Solved Example 3 – Pin fin (insulated tip)

Al rod D=10 mm, L=100 mm, k=200 W/mK, h=20 W/m²K, $T_b$=120°C, $T_\infty$=30°C.

$m = \sqrt{(4h/kD)} = \sqrt{(4\times20/(200\times0.01))} = \sqrt{40} = 6.325\ m^{-1}$; $mL = 0.6325$; $\tanh mL = 0.560$

$\sqrt{(hPkA_c)} = \sqrt{(20\times0.0314\times200\times7.854\times10^{-5})} = 0.0993\ W/K$

$Q = 0.0993 \times 90 \times 0.560 = 5.0\ W$; $Q_{max} = hPL\theta_b = 5.65\ W$; $\eta = 0.885$

## Solved Example 3B – Rectangular fin (convecting tip)

Aluminium fin, thickness t = 3 mm, length L = 40 mm, width w = 1 m, k = 200, h = 40 W/m²K, $T_b$ = 100°C, $T_\infty$ = 30°C. Use corrected length.

$L_c = L + t/2 = 41.5$ mm; $P \approx 2w = 2.006$ m; $A_c = wt = 0.003\ m^2$; $m = \sqrt{(hP/kA_c)} = \sqrt{133.7} = 11.56\ m^{-1}$

$mL_c = 0.480$; $\tanh mL_c = 0.446$; $\sqrt{(hPkA_c)} = \sqrt{(40\times2.006\times200\times0.003)} = 6.94\ W/K$

$Q = 6.94 \times 70 \times 0.446 = 217\ W$; $\eta = \tanh mL_c / mL_c = 0.93$; $\varepsilon = Q/(hA_c\theta_b) = 217/(40\times0.003\times70) = 25.8$

## Use of fins – temperature measurement

A thermometer well (hollow tube) in a pipe behaves like a fin with insulated tip. The sensing tip reads $T_L = T_\infty + \theta_b/\cosh mL$, so error = $(T_L - T_\infty)$ is minimised by large $mL$: long, thin, low-k well and high h.

→ Hence: error ∝ $1/\cosh mL$ → use long well, thin walls, material of low k (stainless steel), insulate the outside of pipe

→ Fin array (n fins): $Q_{total} = n\, Q_{fin} + h\, A_{unfinned}\, \theta_b$; overall surface effectiveness $\eta_o = 1 - (n\, A_f/A_t)(1 - \eta_f)$

Note: Fin design rule: fins are useful when h is small (gases, natural convection). On a water side (h large) fins give little benefit; and $\varepsilon$ is the figure of merit while $\eta$ tells about material utilisation.

# Unsteady (Transient) Heat Conduction

Temperature varies with time as well as position: $\partial T/\partial t \neq 0$. Occurs in heating/cooling of billets, quenching, start-up and shut-down of equipment, thermocouple response.

## Lumped System Analysis

Assume the body is at a uniform temperature at every instant (internal conduction resistance negligible compared with surface convection). Valid when the Biot number is small:

$$Bi = h\, L_c / k = (\text{internal resistance})/(\text{external resistance}) < 0.1\ ,\ L_c = V/A_s \quad \text{criterion}$$

→ $L_c$: sphere R/3, long cylinder R/2, cube a/6, plate of thickness 2L → L (both faces exposed)

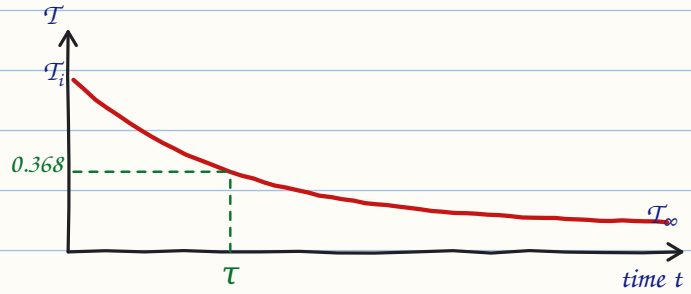
Body ($\rho$, V, c, $T_i$) suddenly placed in fluid at $T_\infty$. Energy balance: $-hA_s(T - T_\infty) = \rho\, V\, c\, dT/dt$. With $\theta = T - T_\infty$ and $\theta_i = T_i - T_\infty$:

$$\theta/\theta_i = (T - T_\infty)/(T_i - T_\infty) = \exp(-hA_s t/\rho V c) = \exp(-Bi \cdot Fo) \quad \text{lumped solution}$$

→ $Fo = \alpha\, t / L_c^2$ (Fourier number, dimensionless time)

→ Time constant $\tau = \rho V c / hA_s = R_{conv} \cdot C_{thermal}$ ; at $t = \tau$, $\theta/\theta_i = e^{-1} = 0.368$ (63.2% of change done)

→ Total heat transferred in time t : $Q = \rho\, V\, c\, (T_i - T_\infty)\, [\, 1 - e^{-t/\tau}\, ]$




Exponential decay of θ

Small τ → fast response

(small mass / large h)

## Solved Example 4 – Cooling of a steel ball

Ball D=10 mm, ρ=7800 kg/m³, c=470 J/kgK, k=45; cools from 400°C in air at 30°C with h=50 W/m²K to 100°C. Find time.

$L_c = D/6 = 1.667\times10^{-3}$ m; $Bi = 50(1.667\times10^{-3})/45 = 0.0019 < 0.1$ → lumped OK

$\tau = \rho\, c\, L_c/h = 7800\times470\times1.667\times10^{-3}/50 = 122$ s; $(100-30)/(400-30) = 0.189 = e^{-t/\tau}$

$t = \tau \ln(1/0.189) = 122 \times 1.665 =$ **203 s**

## Heisler (Transient-Temperature) Charts – Bi > 0.1

When Bi > 0.1, temperature varies inside the body. For a plane wall (thickness 2L), long cylinder or sphere (radius $r_o$) with uniform $T_i$ suddenly exposed to fluid $T_\infty$ (convection), the exact series solution is plotted as Heisler charts using:

$$Bi = h\,L\,/\,k \text{ (or } 1/Bi = k/hL)$$

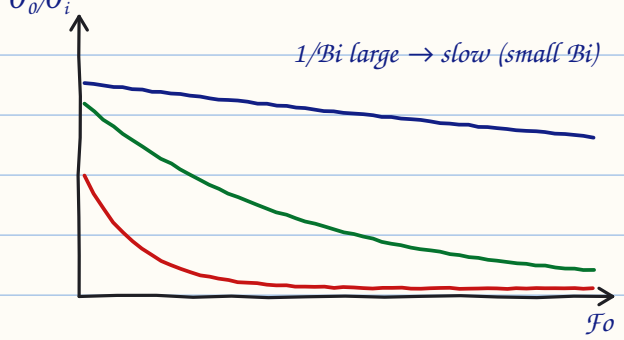
$$Fo = \alpha\, t\,/\,L^2 \text{ (L = half thickness, or } r_o)$$

$$\theta^* = (T - T_\infty)/(T_i - T_\infty)$$

→ Chart 1 : centre-line temperature $\theta_0/\theta_i$ vs Fo for various 1/Bi (x = 0 or r = 0)

→ Chart 2 : position correction $\theta/\theta_0$ vs 1/Bi for various x/L (or $r/r_o$); independent of time

→ Chart 3 : heat loss ratio $Q/Q_0$ vs $Bi^2Fo$ for different Bi; $Q_0 = \rho V c(T_i - T_\infty)$




Centre temp. $\theta_0 = T_0 - T_\infty$

Then $\theta = \theta_0 \times$ (chart 2)

$Q/Q_0$ from chart 3

One-term fit (Fo > 0.2):

$\theta_0/\theta_i = C_1\, e^{-\zeta_1^2 Fo}$

One-term approximation (Fo > 0.2) for a plane wall: $\theta_0/\theta_i = C_1 \exp(-\zeta_1^2 Fo)$, $\zeta_1 \tan\zeta_1 = Bi$, $C_1 = 4\sin\zeta_1/(2\zeta_1 + \sin 2\zeta_1)$. Example: Bi = 1 → $\zeta_1$=0.860, $C_1$=1.119; for Fo = 0.5, $\theta_0/\theta_i = 1.119\, e^{-0.37} =$ **0.773**.

## Procedure for a Heisler-chart problem

1. Compute $Bi = hL/k$. If $Bi < 0.1$ use the lumped method, else Heisler.
2. $Fo = \alpha t/L^2$; read $\theta_0/\theta_i$ from chart 1 → get $T_0$.
3. For a point at $x/L$, read $\theta/\theta_0$ from chart 2; $T = T_\infty + (\theta/\theta_0)(\theta_0/\theta_i)\theta_i$.
4. For total heat, use $Bi^2 Fo$ in chart 3 → $Q = Q_0(Q/Q_0)$.

Extra: Semi-infinite solid (thick body, early times): for surface at $T_s$ suddenly imposed, $(T - T_s)/(T_i - T_s) = erf(x / 2\sqrt{(\alpha t)})$. $Fo < 0.2$ → use this, not the one-term formula.

## Multi-dimensional transient – product solution

Temperature in a short cylinder, rectangular bar or semi-infinite plate is the product of 1-D solutions:

$$\theta^*_{short\ cyl}(r,x,t) = \theta^*_{plane\ wall}(x,t) \times \theta^*_{infinite\ cyl}(r,t)$$

$$\theta^*_{bar}(x,y,t) = \theta^*_{slab\ 1}(x,t) \times \theta^*_{slab\ 2}(y,t)$$ Newman product rule

→ Each 1-D factor is read from the Heisler chart using its own Bi and Fo (own half-thickness / radius)

## Significance of dimensionless numbers

| Number | Definition | Meaning |
|---|---|---|
| Biot Bi | $hL_c/k$ | internal / surface resistance |
| Fourier Fo | $\alpha t/L_c^2$ | conduction rate / storage rate |
| Thermal time constant τ | $\rho Vc/hA$ | time for 63.2% response |

## Solved Example 5 – Is lumped analysis valid?

Aluminium cube, side 5 cm ($k = 205$, $\rho = 2700$, $c = 900$), dropped in water with $h = 500$ $W/m^2K$.

$L_c = a/6 = 8.33$ mm; $Bi = 500(0.00833)/205 = 0.0203 < 0.1$ → lumped valid. $\tau = \rho c L_c/h = 2700\times900\times0.00833/500 = 40.5$ s

For the same cube with $h = 5000$: $Bi = 0.203 > 0.1$ → use Heisler charts.

## Solved Example 6 – Heisler procedure (plane wall)

Large steel plate $2L = 10$ cm ($k = 45$, $\alpha = 1.2\times10^{-5}$ $m^2/s$) at 500°C quenched in oil at 50°C, $h = 450$ $W/m^2K$. Find the mid-plane temperature after 3 min.

$L = 0.05$ m; $Bi = hL/k = 450(0.05)/45 = 0.5$; $Fo = \alpha t/L^2 = 1.2\times10^{-5}(180)/0.0025 = 0.864$ (> 0.2, one-term OK)

For $Bi = 0.5$: $\zeta_1 = 0.6533$, $C_1 = 1.0701$. $\theta_0/\theta_i = 1.0701\ exp(-0.4268 \times 0.864) =$

$1.0701 \times 0.6915 = 0.740$

$T_0 = 50 + 0.740(500 - 50) = 383°C$

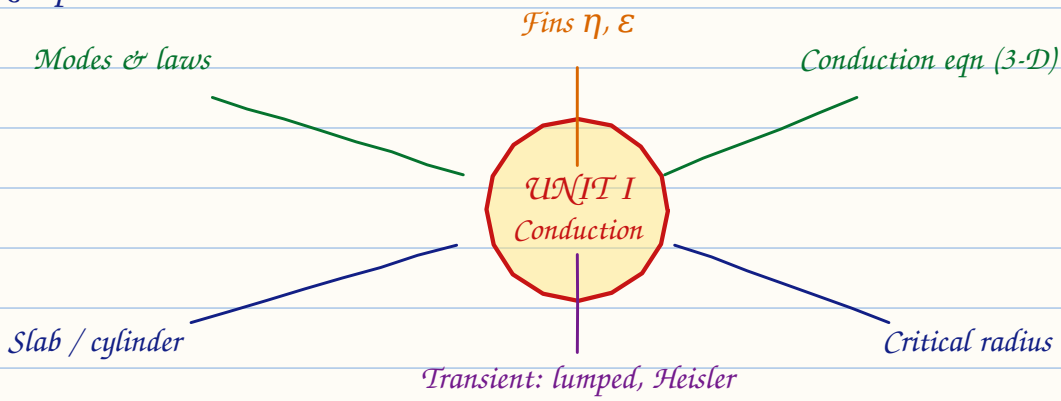
## Important Questions (exam oriented)

### Short answer (2 marks):

1. Define the three modes of heat transfer. State Fourier's law.
2. What is thermal diffusivity? Why is it important in transient conduction?
3. Define critical radius of insulation. Give its expression for a cylinder and a sphere.
4. Define fin efficiency and fin effectiveness. When is a fin justified?
5. What is the Biot number? State the condition for the lumped-capacity method.

### Long answer (10 marks):

1. Derive the general 3-D heat conduction equation in Cartesian co-ordinates. Reduce it to Fourier, Poisson and Laplace equations.
2. Derive an expression for the heat flow through a composite cylinder with convection on both sides.
3. Derive the critical radius of insulation for a cylinder. Explain its significance.
4. Derive the temperature distribution and heat dissipation of a fin with insulated tip.
5. Derive the lumped-system temperature-time relation. Explain Heisler charts with an example.




## Quick Revision – Formulae to remember

| Item | Formula |
|---|---|
| Plane wall | $Q = kA\Delta T/L$ ; $R = L/kA$ |
| Cylinder | $Q = 2\pi kL\Delta T / \ln(r_2/r_1)$ |
| Sphere | $Q = 4\pi k\Delta T / (1/r_1 - 1/r_2)$ |
| Critical radius | cyl: $k/h_o$ sph: $2k/h_o$ |
| Fin (insulated tip) | $Q = \sqrt{(hPkA)}\,\theta_b \tanh mL$ |
| Lumped system | $\theta/\theta_i = e^{-t/\tau}$ , $Bi < 0.1$ |

# UNIT II : Convection

Topics: convection basics, h, free & forced convection, boundary layers, empirical relations, Reynolds–Colburn analogy, integral solution, internal flow

## 2.1 Basic Concepts of Convection

Convection is heat transfer between a solid surface and a moving fluid. It combines conduction (molecular diffusion at the wall) and advection (bulk motion of fluid).

Newton's law of cooling gives the rate:

$$Q = h\,A\,(T_s - T_\infty)\quad q = h\,(T_s - T_\infty)$$ Newton

→ Local coefficient $h_x$ varies along the surface; average $h = (1/L)\int h_x\,dx$

→ At the wall fluid velocity = 0 (no-slip), so heat leaves by pure conduction: $h = -k_f\,(\partial T/\partial y)_{y=0} / (T_s - T_\infty)$

→ h is NOT a property – it depends on velocity V, $\rho$, $\mu$, $c_p$, k, geometry L, surface roughness, flow type

## Types of convection

| Type | Cause of motion | Example | h ($W/m^2K$) |
|---|---|---|---|
| Forced | fan, pump, wind | radiator, heat exchanger | 25 – 250 (air), 50 – 20000 (water) |
| Natural (free) | density difference (buoyancy) | hot plate in still air | 2 – 25 (air) |
| Mixed | both comparable ($Gr/Re^2 \approx 1$) | slow flow in heated pipe | — |

→ Also : laminar vs turbulent; external (plate, cylinder) vs internal (pipe, duct); with or without phase change

## Dimensionless numbers

| Number | Expression | Physical meaning |
|---|---|---|
| Reynolds Re | $\rho V L/\mu = VL/\nu$ | inertia force / viscous force |
| Prandtl Pr | $\mu c_p/k = \nu/\alpha$ | momentum diffusivity / thermal diffusivity |
| Nusselt Nu | $hL/k$ | convection / conduction in fluid layer |
| Grashof Gr | $g\beta\Delta T L^3/\nu^2$ | buoyancy force / viscous force |
| Rayleigh Ra | $Gr \cdot Pr$ | free convection driving strength |
| Stanton St | $h/\rho V c_p = Nu/(Re\ Pr)$ | heat transferred / heat capacity of flow |
| Peclet Pe | $Re \cdot Pr$ | advection / diffusion of heat |

Dimensional analysis (Buckingham $\pi$): Forced : $Nu = f(Re, Pr)$ Free : $Nu = f(Gr, Pr) = f(Ra, Pr)$ (usually $Nu = C\ Re^m\ Pr^n$).

Note: Fluid properties are evaluated at the film temperature $T_f = (T_s + T_\infty)/2$ (unless stated otherwise). For gases, volume expansion coefficient $\beta = 1/T_f$ (Kelvin).

## Dimensional analysis – why Nu = f(Re, Pr)

Forced convection depends on 7 variables : $h, \rho, V, L, \mu, c_p, k$. Fundamental dimensions $M, L, T, \theta = 4$. By Buckingham $\pi$-theorem number of dimensionless groups = 7 - 4 = 3. Choosing repeating variables ($\rho, V, L, k$) :

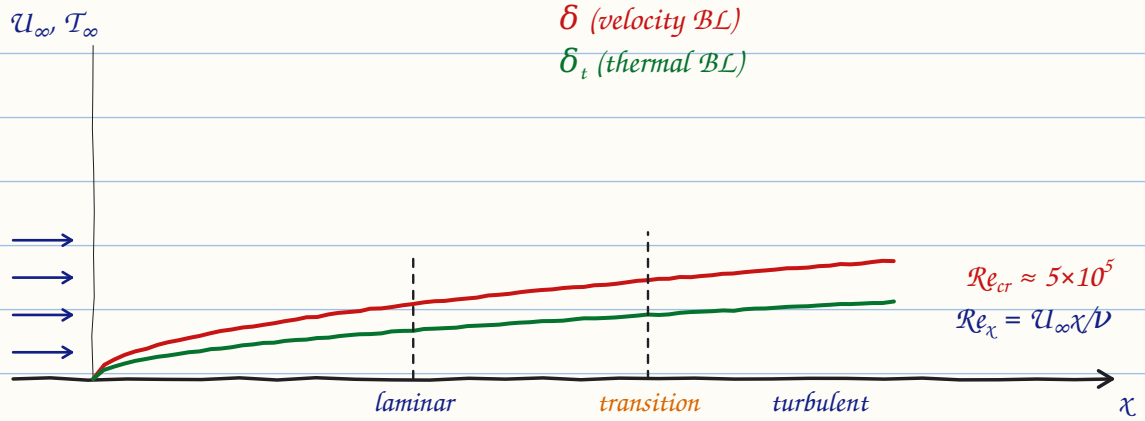
$$\pi_1 = hL/k = Nu\ ,\ \pi_2 = \rho VL/\mu = Re\ ,\ \pi_3 = \mu c_p/k = Pr$$

Hence $Nu = f(Re, Pr)$. For natural convection $V$ is replaced by the buoyancy term $g\beta\Delta T$ and the group $(\rho^2 g\beta\Delta T L^3/\mu^2) = Gr$ appears, so $Nu = f(Gr, Pr)$.

→ The experimental constants $C, m, n$ are then found by plotting log Nu against log Re (or log Ra): a straight line of slope $m$

# Boundary Layer Concepts (External Flow)

Uniform stream $U_\infty$ meets a flat plate at temperature $T_s$. Fluid sticks to the wall; velocity rises from 0 to $U_\infty$ over a thin region:




→ *Hydrodynamic BL thickness* $\delta$ : *distance from wall where* $u = 0.99\,U_\infty$; *wall shear* $\tau_w = \mu\,(\partial u/\partial y)_0$, $C_f = 2\tau_w/\rho U_\infty^2$

→ *Thermal BL thickness* $\delta_t$ : *where* $(T_s - T)/(T_s - T_\infty) = 0.99$; *all heat transfer happens inside* $\delta_t$

→ *Ratio :* $\delta_t/\delta \approx Pr^{-1/3}$ → *liquid metals* $(Pr \ll 1)\ \delta_t \gg \delta$ ; *oils* $(Pr \gg 1)\ \delta_t \ll \delta$ ; *gases* $Pr \approx 0.7 \rightarrow \delta \approx \delta_t$

→ *Laminar → turbulent at* $Re_{cr} \approx 5\times10^5$ *for a flat plate; turbulent BL is thicker, gives larger h and larger drag*

## Boundary-layer equations (steady, 2-D, laminar, constant properties)

$$\partial u/\partial x + \partial v/\partial y = 0 \text{ (continuity)}$$

$$u\,\partial u/\partial x + v\,\partial u/\partial y = \nu\,\partial^2 u/\partial y^2 \text{ (momentum)}$$

$$u\,\partial T/\partial x + v\,\partial T/\partial y = \alpha\,\partial^2 T/\partial y^2 \text{ (energy)}$$

*Similarity (Blasius) solution of these gives the exact laminar results quoted below. When* $Pr = 1$ *the momentum and energy equations are identical in form, so velocity and temperature profiles coincide.*

# Approximate (Integral) Solution – Laminar BL

*Von Karman momentum integral equation for a flat plate* $(dp/dx = 0)$:

$$\tau_w = \rho\ d/dx \int_0^\delta u\,(U_\infty - u)\,dy$$

*Assume a cubic velocity profile satisfying* $u = 0$ *at* $y = 0$, $u = U_\infty$ *and* $\partial u/\partial y = 0$ *at* $y = \delta$:

$$u/U_\infty = (3/2)(y/\delta) - (1/2)(y/\delta)^3$$

*Then* $\tau_w = 3\mu U_\infty/2\delta$ *and* $\int u(U_\infty - u)dy = (39/280)\,U_\infty^2\,\delta$. *Substituting :* $(39/280)$

$\rho U_\infty^2 \, d\delta/dx = 3\mu U_\infty/2\delta \rightarrow \delta \, d\delta = (140/13)(\nu/U_\infty)\, dx$. Integrating with $\delta(0) = 0$:

$$\delta_x = 4.64\, x/\sqrt{Re_x}$$
$$C_{fx} = 0.646/\sqrt{Re_x} \text{ (exact: } 0.664/\sqrt{Re_x})$$
$$C_{f,avg} = 1.328/\sqrt{Re_L}$$

Thermal integral equation: $d/dx \int_0^{\delta_t} (T_\infty - T)\, u\, dy = \alpha\, (\partial T/\partial y)_0$. With a cubic temperature profile $\theta = (3/2)(y/\delta_t) - (1/2)(y/\delta_t)^3$ ($\theta = (T - T_s)/(T_\infty - T_s)$) and the above $u(y)$:

$$\delta_t/\delta = Pr^{-1/3}/1.026$$
$$Nu_x = h_x x/k = 0.332\, Re_x^{1/2}\, Pr^{1/3}$$
$$Nu_L = 0.664\, Re_L^{1/2}\, Pr^{1/3} \quad (h_{avg} = 2h_L)$$

Note: Valid for $Pr \geq 0.6$, laminar flow ($Re_x < 5\times10^5$), constant wall temperature, properties at film temperature. For constant heat flux surface : $Nu_x = 0.453\, Re_x^{1/2}\, Pr^{1/3}$.

## Turbulent flat plate – 1/7 power-law result

Take $u/U_\infty = (y/\delta)^{1/7}$ and the pipe-flow wall shear $\tau_w = 0.0225\, \rho U_\infty^2 (\nu/U_\infty\delta)^{1/4}$. Then $\int u(U_\infty - u)dy = (7/72)U_\infty^2\delta$. The momentum integral gives:

$$\delta = 0.37\, x\, Re_x^{-1/5}$$
$$C_{f,x} \approx 0.0592\, Re_x^{-1/5}$$
$$C_{f,avg} = 0.074\, Re_L^{-1/5} \quad (5\times10^5 < Re_L < 10^7)$$

→ Turbulent $\delta$ grows as $x^{0.8}$ (faster than laminar $x^{0.5}$); eddies mix the fluid so wall gradients and $h$ are much larger

→ Using Colburn analogy: $Nu_x = (C_{f,x}/2)\, Re_x\, Pr^{1/3} = 0.0296\, Re_x^{0.8}\, Pr^{1/3}$

# Fluid Friction – Heat Transfer Analogy

Momentum and heat transfer in a boundary layer are governed by similar equations, so skin friction and heat transfer coefficient are related:

| Analogy | Relation | Valid for |
|---|---|---|
| Reynolds | $St = C_f/2$ | $Pr = 1$ (laminar & turbulent) |
| Prandtl–Taylor | $St = (C_f/2) / [1 + 5\sqrt{(C_f/2)}(Pr-1)]$ | $Pr \neq 1$ |
| Chilton–Colburn | $St \cdot Pr^{2/3} = C_f/2$ ($j_H = j_f$) | $0.6 < Pr < 60$ |

→ Flat plate : $C_{f,x} = 0.664\ Re_x^{-1/2}$ laminar ; $0.0592\ Re_x^{-1/5}$ turbulent. Hence knowing friction, h is found: $h = (C_f/2)\ \rho\ c_p\ U_\infty\ Pr^{-2/3}$

→ Practical use : drag measured in wind tunnel → heat transfer estimated (and vice-versa); explains why good heat-transfer surfaces create high pressure drop

## Reynolds analogy – quick derivation

Shear stress: $\tau = (\mu + \rho\varepsilon_m)\ du/dy$ ; heat flux: $q = -(k + \rho c_p \varepsilon_h)\ dT/dy$. If $Pr = 1$ and $\varepsilon_m = \varepsilon_h$, then $q/\tau = -c_p\ dT/du$ . Integrating from the wall to the free stream ($\tau \approx \tau_w$, $q \approx q_w$):

$$q_w/\tau_w = c_p(T_s - T_\infty)/U_\infty \rightarrow h/(\rho\ c_p\ U_\infty) = C_f/2 \text{ i.e. } St = C_f/2$$

Colburn extended this to $0.6 < Pr < 60$ by introducing $Pr^{2/3}$. Used chiefly for turbulent flow and for estimating h from a measured pressure drop in pipes.

# Forced Convection – External Flow Correlations

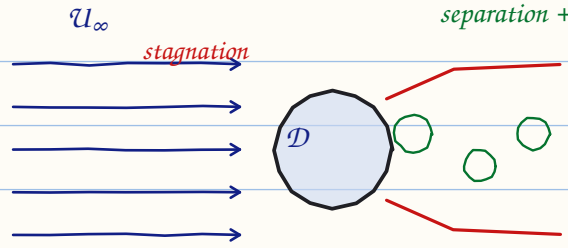
## (a) Flat plate (properties at $T_f$)

| Flow condition | Local $Nu_x$ | Average $Nu_L$ |
|---|---|---|
| Laminar, $Re < 5\times10^5$, $Pr>0.6$ | $0.332\ Re_x^{0.5}\ Pr^{1/3}$ | $0.664\ Re_L^{0.5}\ Pr^{1/3}$ |
| Turbulent, $Re > 5\times10^5$ | $0.0296\ Re_x^{0.8}\ Pr^{1/3}$ | $0.037\ Re_L^{0.8}\ Pr^{1/3}$ |
| Mixed (laminar + turbulent) | — | $(0.037\ Re_L^{0.8} - 871)\ Pr^{1/3}$ |
| Liquid metals, $Pr < 0.05$ | $0.565\ (Re_x\ Pr)^{0.5}$ | $1.13\ (Re_L Pr)^{0.5}$ |

→ Constant heat flux : $Nu_x = 0.453\ Re_x^{1/2}\ Pr^{1/3}$ (laminar), $Nu_x = 0.0308\ Re_x^{0.8}\ Pr^{1/3}$ (turbulent)

→ Heat transfer rate $Q = h_{avg}\ A\ (T_s - T_\infty)$; drag $F = C_{f,avg}\ (\frac{1}{2}\rho U_\infty^2)\ A$

## (b) Cylinder in cross-flow ($Re_D = U_\infty D/\nu$)




Laminar BL : separation at ≈ 80°

Turbulent BL : at ≈ 140° ($Re > 2\times10^5$)

h is max at the front stagnation point, min near separation.

**Hilpert : $Nu_D = C\,Re_D^{\,n}\,Pr^{1/3}$**

| $Re_D$ | 0.4–4 | 4–40 | 40–4000 | 4000–40000 | 40000–400000 |
|---|---|---|---|---|---|
| $C$ | 0.989 | 0.911 | 0.683 | 0.193 | 0.027 |
| $n$ | 0.330 | 0.385 | 0.466 | 0.618 | 0.805 |

Churchill–Bernstein (all $Re\cdot Pr > 0.2$): $Nu_D = 0.3 + 0.62\,Re_D^{1/2}Pr^{1/3} / [1 + (0.4/Pr)^{2/3}]^{1/4} \cdot [1 + (Re_D/282000)^{5/8}]^{4/5}$

→ Sphere (Whitaker) : $Nu_D = 2 + (0.4\,Re_D^{1/2} + 0.06\,Re_D^{2/3})\,Pr^{0.4}\,(\mu_\infty/\mu_s)^{1/4}$ ; $Nu \to 2$ for still fluid

→ Non-circular cylinders / tube banks use the same form with different C, n (Jakob, Zhukauskas); in banks velocity is taken at minimum flow area

## Solved Example 1 – Flat plate, laminar

Air at 20°C flows at 5 m/s over a 1 m long, 1 m wide plate at 80°C. $T_f = 50°C$: $\nu = 1.79\times10^{-5}$, $k = 0.028$ W/mK, $Pr = 0.703$. Find average h and heat loss.

$Re_L = 5(1)/1.79\times10^{-5} = 2.79\times10^5 < 5\times10^5$ → laminar. $Nu_L = 0.664$

$(2.79\times10^5)^{0.5}(0.703)^{1/3} = 0.664 \times 528.5 \times 0.889 = 312$

$h = Nu\,k/L = 312 \times 0.028 / 1 =$ **8.7 W/m²K**; $Q = hA\Delta T = 8.7 \times 1 \times 60 =$ **524 W** (one side)

## Solved Example 2 – Cylinder in cross-flow

Air at 20°C, 10 m/s over a 25 mm tube at 120°C. At $T_f = 70°C$ : $\nu = 1.995\times10^{-5}$, $k = 0.0295$, $Pr = 0.70$. Find h and loss per metre.

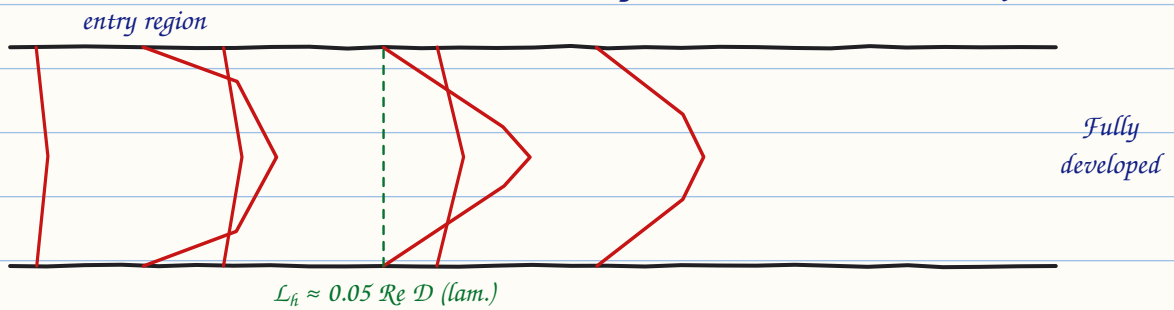
$Re_D = 10(0.025)/1.995\times10^{-5} = 12\,530$ → $C = 0.193$, $n = 0.618$. $Nu = 0.193$

$(12530)^{0.618}(0.7)^{1/3} = 0.193 \times 340 \times 0.888 = 58.3$

$h = 58.3 \times 0.0295/0.025 =$ **68.8 W/m²K**; $Q/L = h\,\pi\,D\,\Delta T = 68.8 \times \pi \times 0.025 \times 100 =$ **540 W/m**

# Internal Flow – Convection in Pipes

Flow in a tube : $Re_D = \rho\, u_m\, D/\mu$ . $Re < 2300$ laminar, 2300–4000 transition, $Re > 4000$ turbulent. For non-circular ducts use hydraulic diameter $D_h = 4A_c/P$.




→ Hydrodynamic entry length : laminar $L_h \approx 0.05\, Re\, D$ ; turbulent $L_h \approx 10\, D$. Thermal entry length : laminar $L_t \approx 0.05\, Re\, Pr\, D$

→ Bulk (mean) temperature $T_b$ replaces $T_\infty$: $Q = h\, A\, (T_s - T_b)$; energy balance $Q = \dot{m}\, c_p\, (T_{out} - T_{in})$

| Case | Correlation (properties at $T_b$) | Range |
|---|---|---|
| Laminar, fully dev., $T_s$ const | $Nu_D = 3.66$ | $Re < 2300$ |
| Laminar, fully dev., $q$ const | $Nu_D = 4.36$ | $Re < 2300$ |
| Turbulent : Dittus–Boelter | $Nu_D = 0.023\, Re^{0.8}\, Pr^n$ ($n$=0.4 heating, 0.3 cooling) | $Re>10^4$, $0.6<Pr<160$, $L/D>10$ |
| Sieder–Tate (viscous liquids) | $Nu_D = 0.027\, Re^{0.8}\, Pr^{1/3}\, (\mu_b/\mu_s)^{0.14}$ | $Re>10^4$ |
| Gnielinski (accurate) | $Nu = (f/8)(Re-1000)Pr \,/\, [1+12.7\sqrt{(f/8)}(Pr^{2/3}-1)]$ | $3000<Re<5\times10^6$ |

Constant wall temperature (e.g. condensing steam outside):

$$(T_s - T_{out}) / (T_s - T_{in}) = \exp(-h\, P\, L / \dot{m}\, c_p)$$

$$Q = \dot{m}\, c_p\, (T_{out} - T_{in}) = h\, A\, \Delta T_{lm},\ \Delta T_{lm} = (\Delta T_{in} - \Delta T_{out}) / \ln(\Delta T_{in}/\Delta T_{out})$$

Constant heat flux (electric heater): $T_b(x) = T_{in} + q\, P\, x/(\dot{m}\, c_p)$ rises linearly; wall temperature $T_s = T_b + q/h$ also rises linearly after the entry length.

## Solved Example 3 – Water heated in a tube

Water at 1 m/s through a 20 mm tube, 30°C in, 50°C out, wall at 80°C. Properties at 40°C: $\rho = 992$, $c_p = 4174$, $\nu = 0.658\times10^{-6}$, $k = 0.63$, $Pr = 4.3$. Find $h$ and length.

$Re = 1(0.02)/0.658\times10^{-6} = 30\,400$ (turbulent). $Nu = 0.023\,(30400)^{0.8}(4.3)^{0.4} = 0.023 \times 3857 \times 1.79 = 159$

$h = 159(0.63)/0.02 = 5010\ W/m^2K$; $\dot{m} = 992(1)(\pi/4)(0.02)^2 = 0.3116\ kg/s$; $Q =$

$0.3116(4174)(20) = 26$ kW

$\ln[(80-50)/(80-30)] = -0.511 = -\,h\pi DL/\dot{m}c_p \rightarrow L = 0.511(1300.7)/(5010 \times 0.0628) = 2.11$ m

## Pressure drop & pumping power (internal flow)

$$\Delta p = f\,(L/D)\,(\rho\, u_m^2/2)$$

$$f = 64/Re \text{ (laminar)}$$

$$f = 0.316\,Re^{-1/4} \text{ (turbulent smooth, } Re < 2\times10^4\text{)};\ f = 0.184\,Re^{-0.2} \text{ (} Re > 2\times10^4\text{)}$$

$$\text{Pumping power} = \dot{m}\,\Delta p/\rho = \mathcal{V}\Delta p$$

Example 3 data: $f = 0.316/(30400)^{0.25} = 0.0239$; $\Delta p = 0.0239\,(2.11/0.02)(992 \times 1^2/2) = 1.25$ kPa; power $= 0.3116 \times 1250/992 = 0.39$ W.

## Solved Example 3B – Air in a rectangular duct

Duct 5 cm × 10 cm, air at 40°C, 5 m/s ($\nu = 1.70\times10^{-5}$, $k = 0.0265$, $Pr = 0.70$), being heated.

$D_h = 4A/P = 4(0.05 \times 0.10)/(2 \times 0.15) = 0.0667$ m; $Re = 5(0.0667)/1.70\times10^{-5} = 19\,600$ (turbulent)

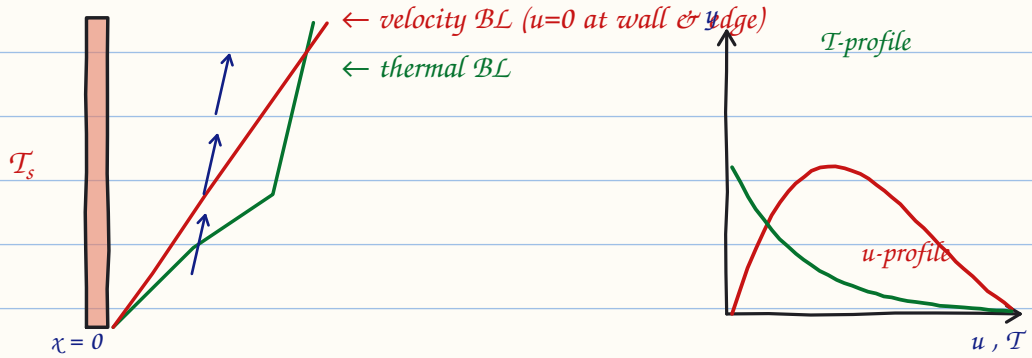
$Nu = 0.023\,(19600)^{0.8}(0.7)^{0.4} = 0.023 \times 2723 \times 0.867 = 54.3$; $h = 54.3(0.0265)/0.0667 = 21.6$ W/m²K

## Solved Example 3C – Laminar flow of oil

Oil ($k = 0.14$ W/mK) at $Re = 500$ in a 10 mm tube with uniform wall heat flux. $Re < 2300 \rightarrow$ laminar, fully developed $Nu = 4.36 \Rightarrow h = 4.36(0.14)/0.01 = 61$ W/m²K (independent of velocity!). Thermal entry length $L_t = 0.05\,Re\,Pr\,D$ grows with Pr: for $Pr = 100$, $L_t = 0.05(500)(100)(0.01) = 25$ m.

# Free (Natural) Convection

Fluid motion caused by buoyancy: heating reduces density, the lighter fluid near the wall rises. No external pump/fan; velocity is zero at the wall AND zero far away, with a maximum inside the boundary layer.




→ Boussinesq : $\rho$ varies only in the buoyancy term; $\beta = -(1/\rho)(\partial\rho/\partial T)_p \rightarrow \beta = 1/T$ (K) for ideal gas

→ Governing parameter : $Gr = g\beta (T_s - T_\infty) L^3/\nu^2$, $Ra = Gr \cdot Pr$. Vertical plate: laminar $Ra < 10^9$, turbulent $Ra > 10^9$

→ Exact (similarity) solution, vertical plate: $Nu_x = 0.508\, Pr^{1/2} (0.952 + Pr)^{1/4}\, Gr_x^{1/4}$

## Integral analysis of the free-convection BL (vertical plate)

Assume cubic-type profiles $T - T_\infty = (T_s - T_\infty)(1 - y/\delta)^2$ and $u = u_x (y/\delta)(1 - y/\delta)^2$.

Momentum balance (inertia = viscous + buoyancy):

$$d/dx \int_0^\delta u^2\, dy = -\nu\, (du/dy)_0 + g\beta \int_0^\delta (T - T_\infty)\, dy$$

Solving the momentum and energy integrals together gives (Squire–Eckert result):

$$\delta/x = 3.93\, Pr^{-1/2} (0.952 + Pr)^{1/4}\, Gr_x^{-1/4}$$

$$Nu_x = 0.508\, Pr^{1/2} (0.952 + Pr)^{1/4}\, Gr_x^{1/4}$$

→ $\delta \propto x^{1/4}$, $h_x \propto x^{-1/4}$ → average $h$ over height $L$ is $h_{avg} = (4/3)\, h_{x=L}$

## Empirical correlations : $Nu = h L/k = C (Gr\, Pr)^n = C\, Ra^n$ (properties at $T_f$)

| Geometry (characteristic L) | Range of Ra | C | n |
|---|---|---|---|
| Vertical plate / cylinder (height L) | $10^4 - 10^9$ | 0.59 | 1/4 |
| Vertical plate / cylinder | $10^9 - 10^{13}$ | 0.10 | 1/3 |
| Horizontal plate, hot face UP (L = A/P) | $10^4 - 10^7$ | 0.54 | 1/4 |
| Horizontal plate, hot face UP | $10^7 - 10^{11}$ | 0.15 | 1/3 |
| Horizontal plate, hot face DOWN | $10^5 - 10^{11}$ | 0.27 | 1/4 |
| Horizontal cylinder (L = D) | $10^4 - 10^7$ | 0.48 | 1/4 |
| Horizontal cylinder (L = D) | $10^7 - 10^{12}$ | 0.125 | 1/3 |
| Sphere | $Ra < 10^{11}$ | $Nu = 2 + 0.43\, Ra^{1/4}$ | |

Churchill–Chu (vertical, all Ra): $Nu = \{ 0.825 + 0.387\, Ra^{1/6} / [1 + (0.492/Pr)^{9/16}]^{8/27} \}^2$

Horizontal cylinder ($Ra_D < 10^{12}$): $Nu = \{ 0.6 + 0.387\, Ra_D^{1/6} / [1 + (0.559/Pr)^{9/16}]^{8/27} \}^2$

Note: A vertical cylinder behaves like a vertical plate if $D \geq 35\, L / Gr_L^{1/4}$. For a plate inclined at $\theta$ to the vertical replace $g$ by $g \cos\theta$ ($\theta < 60°$).

## Solved Example 4 – Vertical plate

Vertical plate 0.5 m high at 90°C in still air at 30°C ($T_f = 60°C$ : $\nu = 1.896\times10^{-5}$, $k = 0.0287$, $Pr = 0.70$, $\beta = 1/333\ K^{-1}$). Find $h$ and heat loss per metre width.

$Ra = g\beta\Delta T L^3 Pr/\nu^2 = 9.81(0.003)(60)(0.125)(0.7)/(3.595\times10^{-10}) = 4.3\times10^8 < 10^9 \rightarrow$ laminar

$Nu = 0.59\ (4.3\times10^8)^{1/4} = 0.59 \times 144 = 85$; $h = 85(0.0287)/0.5 =$ **4.9 W/m²K**; $Q = hL\Delta T = 4.9(0.5)(60) =$ **146 W/m**

## Solved Example 5 – Horizontal pipe

Pipe $D = 0.1$ m at 80°C in air at 20°C ($T_f = 50°C$: $\nu = 1.79\times10^{-5}$, $k = 0.028$, $Pr = 0.703$, $\beta = 1/323$). $Ra_D = 9.81(1/323)(60)(0.001)(0.703)/(3.2\times10^{-10}) = 4.0\times10^6$

$Nu = 0.48\, Ra^{1/4} = 0.48(44.7) = 21.5$; $h = 21.5(0.028)/0.1 =$ **6.0 W/m²K**; $Q/L = h\,\pi\, D\,\Delta T = 6.0\,\pi\,(0.1)(60) =$ **113 W/m**

## Solved Example 6 – Sphere in still air

Sphere $D = 0.2$ m at 100°C in air at 20°C ($T_f = 60°C$, $\nu = 1.896\times10^{-5}$, $k = 0.0287$, $Pr = 0.7$, $\beta = 1/333$). $Ra_D = 9.81(0.003)(80)(0.2)^3(0.7)/(3.595\times10^{-10}) = 3.7\times10^7$

$Nu = 2 + 0.43\, Ra^{1/4} = 2 + 0.43(77.8) = 35.5$; $h = 35.5(0.0287)/0.2 =$ **5.1 W/m²K**; $Q = h\,\pi D^2\,\Delta T = 5.1(0.1257)(80) =$ **51 W**

## Which correlation? – quick decision guide

| Situation | Use |
|---|---|
| Plate, flow along it | $Re_L < 5\times10^5$: $0.664Re^{1/2}Pr^{1/3}$; else $(0.037Re^{0.8}-871)Pr^{1/3}$ |
| Cylinder, cross flow | Hilpert $Nu = C\,Re^n\,Pr^{1/3}$ or Churchill–Bernstein |
| Sphere, forced | Whitaker $Nu = 2 + (0.4Re^{1/2}+0.06Re^{2/3})Pr^{0.4}$ |
| Pipe, laminar | $Nu = 3.66$ ($T_s$) or $4.36$ ($q$); $Re<2300$ |
| Pipe, turbulent | Dittus–Boelter / Sieder–Tate / Gnielinski |
| Free convection | $Nu = C\,Ra^n$; choose $C,n$ by geometry & $Ra$ |
| Mixed convection | $Gr/Re^2 \approx 1$ : $Nu^3 = Nu_{forced}^{\,3} \pm Nu_{free}^{\,3}$ |

Remember: $Gr/Re^2 \gg 1$ → forced convection dominates ; $Gr/Re^2 \ll 1$ → free convection dominates ; $\approx 1$ → mixed. Properties of fluid: evaluate at $T_f$; in pipes at bulk mean temperature.

## Solved Example 7 – Plate with mixed boundary layer

Air at 20°C, $U = 30$ m/s flows over a 2 m long plate at 60°C ($T_f = 40$°C: $\nu = 1.70\times10^{-5}$, $k = 0.0265$, $Pr = 0.70$). Find $h_{avg}$ and heat loss per metre width.

$Re_L = 30(2)/1.70\times10^{-5} = 3.53\times10^6 > 5\times10^5$; transition at $x_c = 5\times10^5(1.7\times10^{-5})/30 = 0.28$ m → mixed BL

$Nu_L = (0.037\,Re^{0.8} - 871)Pr^{1/3} = (0.037 \times 1.73\times10^5 - 871)(0.888) = 4915$; $h = 4915(0.0265)/2 =$ **65 W/m²K**

$Q = hL\Delta T(1\text{ m}) = 65(2)(40) =$ **5.2 kW per metre width**

## Solved Example 8 – Hot horizontal plate facing up

Square plate 0.3 m × 0.3 m at 80°C in air at 20°C ($T_f = 50$°C; $\nu = 1.79\times10^{-5}$, $k = 0.028$, $Pr = 0.703$, $\beta = 1/323$). $L_c = A/P = 0.09/1.2 = 0.075$ m.

$Ra = 9.81(1/323)(60)(0.075)^3(0.703)/(3.2\times10^{-10}) = 1.7\times10^6$; $Nu = 0.54\,Ra^{1/4} = 0.54(36.0) = 19.5$; $h = 19.5(0.028)/0.075 = 7.3$

$Q_{top} = 7.3(0.09)(60) =$ **39 W**. (If the hot face were DOWN: $Nu = 0.27\,Ra^{1/4}$ → half the loss.)

## Laminar vs Turbulent – comparison

| Aspect | Laminar | Turbulent |
|---|---|---|
| Flow pattern | smooth layers | eddies, mixing |
| BL thickness $\delta$ | $\propto x^{1/2}$ | $\propto x^{4/5}$ |
| $Nu_x \propto$ | $Re^{1/2}$ | $Re^{0.8}$ |
| $h$, $C_f$ | low | high |
| Where | $Re_x < 5\times10^5$ / $Re_D < 2300$ | $Re_x > 5\times10^5$ / $Re_D > 4000$ |

## Properties of water (approx.)

| $T$ (°C) | $\rho$ (kg/m$^3$) | $c_p$ (J/kgK) | $k$ (W/mK) | $\nu$ (m$^2$/s) | Pr |
|---|---|---|---|---|---|
| 20 | 998 | 4182 | 0.598 | $1.00\times10^{-6}$ | 7.0 |
| 40 | 992 | 4179 | 0.631 | $0.66\times10^{-6}$ | 4.3 |
| 60 | 983 | 4185 | 0.654 | $0.48\times10^{-6}$ | 3.0 |
| 80 | 972 | 4197 | 0.670 | $0.36\times10^{-6}$ | 2.2 |

## Important Questions (exam oriented)

1. Distinguish free and forced convection. Define h, Nu, Pr, Gr and Ra with physical significance.
2. Explain hydrodynamic and thermal boundary layers on a flat plate with neat sketches.
3. Obtain $\delta$, $C_f$ and $Nu_x$ for laminar flow over a flat plate using the integral method.
4. Explain Reynolds–Colburn analogy and its limitations.
5. Explain development of the boundary layer on a heated vertical plate. State correlations for plates & cylinders.
6. Write Dittus–Boelter equation. Explain thermal and hydrodynamic entry lengths in pipes.

## Quick viva (short answers)

Q1.Why is h not a property? – it depends on flow, geometry and fluid, unlike k.

Q2.Why is the gas-side heat transfer coefficient low? – low k and $\rho$ of gas → fins on gas side.

Q3.Pr of liquid metals? – □ 1 (0.01–0.03): thermal BL thicker than velocity BL.

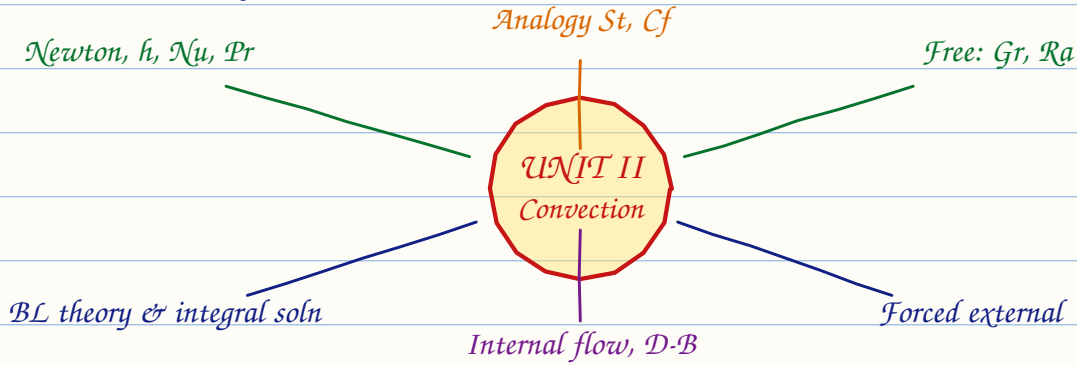
Q4.Why does turbulence raise h? – eddy mixing carries energy across the layer, thinning

the sub-layer.

Q5.When does the flat-plate BL become turbulent? – $Re_x \approx 5\times10^5$.

Q6.Which temperature is used for properties? – film temperature; bulk mean temperature for pipes.

Q7.What is the role of Gr in natural convection? – it replaces Re (buoyancy/viscous).



# UNIT III : Boiling and Condensation

Topics: pool boiling regimes, boiling curve, bubble physics, Rohsenow / Zuber / Bromley, flow boiling, film & dropwise condensation, Nusselt theory, problems

## 3.1 Introduction – Phase-Change Heat Transfer

When a fluid changes phase at a surface (liquid → vapour = boiling, vapour → liquid = condensation) large amounts of latent heat $h_{fg}$ are exchanged at almost constant temperature. Hence h is very high ($10^3 – 10^5$ $W/m^2K$) compared with single-phase convection.

→ Boiling occurs when a surface is held above the saturation temperature : excess temperature $\Delta T_e = T_s - T_{sat}$

→ Evaporation = liquid-vapour change at the free surface with no bubbles ($T_s < T_{sat}$); boiling = bubbles form at the heated solid surface

→ Newton's law is still used : $q = h\,(T_s - T_{sat}) = h\,\Delta T_e$ for boiling, $q = h\,(T_{sat} - T_w)$ for condensation

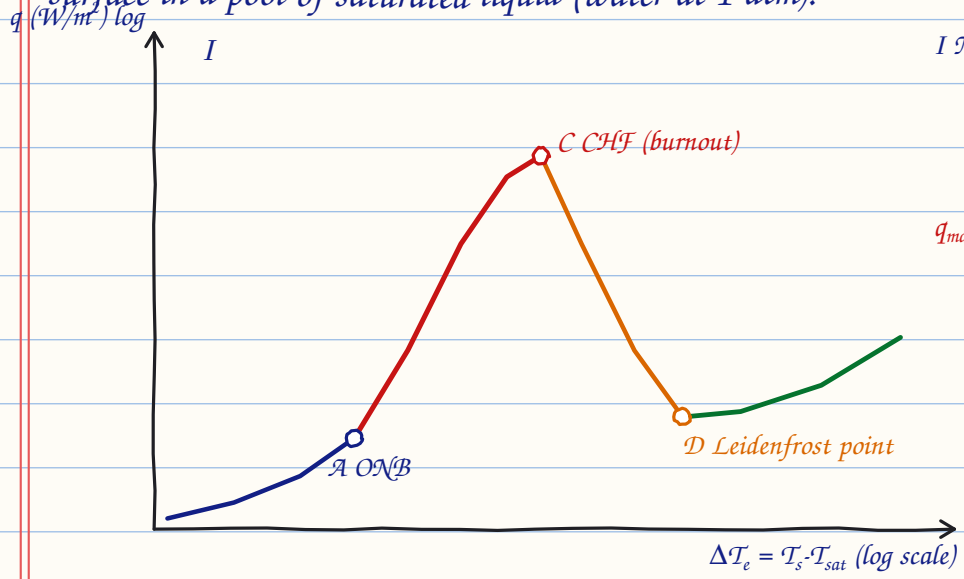
| Process | Typical h ($W/m^2K$) |
|---|---|
| Free convection, air | 5 – 25 |
| Forced convection, water | 500 – 10 000 |
| Nucleate boiling of water | 2 500 – 35 000 (up to $10^5$) |
| Film condensation, steam | 4 000 – 11 000 |
| Dropwise condensation, steam | 20 000 – 250 000 |
| Film boiling | 100 – 400 |

## Classification of boiling

| Basis | Types |
|---|---|
| Fluid motion | Pool boiling (stagnant liquid, motion by buoyancy only) \| Flow boiling (forced flow in tube/duct) |
| Liquid bulk temperature | Saturated boiling ($T_{liq} = T_{sat}$) \| Sub-cooled boiling ($T_{liq} < T_{sat}$, bubbles collapse in bulk) |

# Pool Boiling Curve (Nukiyama, 1934)

Heat flux $q$ is plotted against excess temperature $\Delta T_e$ on a log-log graph for a heated surface in a pool of saturated liquid (water at 1 atm):




I Natural convection ($\Delta T_e$ < 5°C)
II Nucleate boiling (5–30°C)
III Transition / partial film (30–120°C)
IV Stable film boiling (> 120°C)

$q_{max} \approx 1.1\ MW/m^2$ at $\Delta T_e \approx 30°C$
$q_{min} \approx 0.02\ MW/m^2$ (Leidenfrost)

## Regimes in detail

→ I – Natural-convection boiling ($\Delta T_e \leq 5°C$): no bubbles; liquid superheated slightly, evaporation at free surface; h as free convection.

→ II – Nucleate boiling (5–30°C): bubbles form at nucleation sites. IIa isolated bubbles (rise, collapse in sub-cooled liquid); IIb slugs and columns (bubbles merge, jets). Intense stirring → very high h; h and q increase rapidly with $\Delta T_e$ ($q \propto \Delta T_e^3$). The most desirable regime – operate here.

→ Point C – Critical (peak) heat flux $q_{max}$, "burnout / DNB" : vapour escapes so fast that liquid cannot reach the surface; a vapour blanket starts forming.

→ III – Transition (partial film) boiling (30–120°C): unstable alternating film + nucleate; q falls as $\Delta T_e$ rises because vapour film ($k_v$ small) insulates the surface.

→ Point D – Leidenfrost point : minimum heat flux $q_{min}$ where a stable vapour film is first established.

→ IV – Stable film boiling ($\Delta T_e$ > 120°C): surface covered by vapour film; heat by conduction + radiation through the film; q rises again, but h is low and surface becomes very hot.

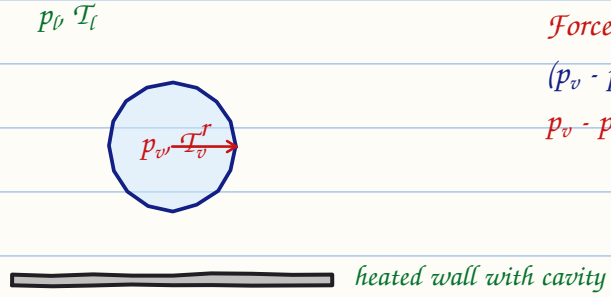
Important: Burnout danger : in heat-flux controlled systems (electric heaters, nuclear fuel rods) when $q$ exceeds $q_{max}$ the surface temperature jumps from ≈ 130°C to > 1000°C (point C → film boiling) and the heater may melt. In temperature-controlled systems (steam heating) the curve can be followed through the transition region.

## Dimensionless groups of boiling

| Group | Definition | Meaning |
|---|---|---|
| Jakob Ja | $c_{pl}\,\Delta T_e / h_{fg}$ | sensible heat / latent heat |
| Bond Bo | $g(\rho_l-\rho_v)L^2/\sigma$ | buoyancy / surface tension |
| Boiling no. $Bo_i$ | $q/(G\,h_{fg})$ | heat flux / mass-flux latent heat |

# Bubble Formation and Growth

A vapour bubble of radius r in equilibrium with surrounding liquid needs the vapour pressure to exceed the liquid pressure by surface tension:




Force balance on half-bubble :

$(p_v - p_l)\,\pi r^2 = 2\pi r \sigma$

$p_v - p_l = 2\sigma / r$

Using Clausius–Clapeyron ($dp/dT = h_{fg}\rho_v/T_{sat}$) the superheat needed inside the bubble is:

$$T_v - T_{sat} = 2\,\sigma\,T_{sat} / (\rho_v\,h_{fg}\,r)$$

→ Smaller radius → greater superheat needed. Hence bubbles form at nucleation sites: tiny cavities/scratches that trap gas/vapour (large effective r)

→ Rough surfaces boil at lower $\Delta T_e$; very clean smooth surfaces give large superheat then violent boiling

→ Departure diameter (Fritz) : $D_b = 0.0208\,\theta\,\sqrt{[\,\sigma / g(\rho_l - \rho_v)\,]}$ ($\theta$ = contact angle in degrees)

→ Heat transfer in nucleate boiling is by (i) latent heat carried by bubbles, (ii) liquid micro-layer evaporation, (iii) agitation – the "pumping action" of departing bubbles

# Correlations for Pool Boiling

## (1) Nucleate boiling – Rohsenow (1952)

$$q_{nb} = \mu_l h_{fg} \left[ g(\rho_l - \rho_v) / \sigma \right]^{1/2} \left[ c_{pl} \Delta T_e / \left( C_{sf} h_{fg} Pr_l^n \right) \right]^3 \quad \text{Rohsenow}$$

→ $C_{sf}$ = surface–fluid constant (depends on material & finish); $n = 1.0$ for water, 1.7 for other fluids; properties of saturated liquid

| Surface–fluid | $C_{sf}$ |
|---|---|
| Water – polished copper | 0.0130 |
| Water – scored copper | 0.0068 |
| Water – ground stainless steel | 0.0080 |
| Water – brass | 0.0060 |
| Water – nickel | 0.0060 |
| Water – platinum | 0.0130 |
| n-Pentane – polished copper | 0.0154 |
| Ethanol – chromium | 0.0027 |

→ $q \propto \Delta T_e^3$ → heat transfer coefficient $h = q/\Delta T_e \propto \Delta T_e^2$. Rearranged : $\Delta T_e = (C_{sf} h_{fg} Pr_l^n / c_{pl}) \left[ q/(\mu_l h_{fg}) \sqrt{(\sigma/g(\rho_l - \rho_v))} \right]^{1/3}$

## (2) Peak (critical) heat flux – Zuber

$$q_{max} = 0.131 \, h_{fg} \, \rho_v^{1/2} \left[ \sigma \, g \, (\rho_l - \rho_v) \right]^{1/4} \quad \text{Zuber (large flat plate)}$$

→ For water at 1 atm $q_{max} \approx 1.1$ MW/m$^2$; increases with pressure up to ≈ 1/3 of critical pressure then falls. Independent of surface material!

→ Minimum heat flux (Zuber/Berenson) : $q_{min} = 0.09 \, \rho_v \, h_{fg} \left[ \sigma \, g \, (\rho_l - \rho_v) / (\rho_l + \rho_v)^2 \right]^{1/4}$

## (3) Film boiling – Bromley (horizontal cylinder / sphere)

$$Nu_D = h_{conv} D/k_v = C \left[ g \, \rho_v (\rho_l - \rho_v) \, h'_{fg} \, D^3 / \left( \mu_v \, k_v \, \Delta T_e \right) \right]^{1/4}$$

$$h'_{fg} = h_{fg} + 0.4 \, c_{pv} \, \Delta T_e \; ; \; C = 0.62 \text{ (cylinder)}, \; 0.67 \text{ (sphere)}$$

At high $\Delta T_e$ radiation through the film adds : $h_{rad} = \varepsilon \sigma (T_s^4 - T_{sat}^4)/(T_s - T_{sat})$. Combined coefficient (approximately):

$$h^{4/3} = h_{conv}^{4/3} + h_{rad} \, h^{1/3} \quad (\text{if } h_{rad} < h_{conv} : h \approx h_{conv} + 0.75 \, h_{rad})$$

→ In film boiling properties of vapour are taken at film temperature $T_f = (T_s + T_{sat})/2$

## Solved Example 1 – Nucleate boiling of water

Water at 1 atm (100°C) boils on a polished copper surface kept at 110°C. Saturated liquid : $\rho_l = 957.9$, $\rho_v = 0.6$ kg/m$^3$, $h_{fg} = 2257$ kJ/kg, $\mu_l = 282\times10^{-6}$ Pa·s, $c_{pl} = 4217$ J/kgK, $Pr_l = 1.75$, $\sigma = 0.0589$ N/m, $C_{sf} = 0.013$, $n = 1$.

$\Delta T_e = 10$°C. $\mu_l h_{fg} = 282\times10^{-6}(2.257\times10^6) = 636.5$; $[g(\rho_l-\rho_v)/\sigma]^{1/2} = (9.81 \times 957.3/0.0589)^{1/2} = 399.3$

$[c_{pl}\Delta T_e/(C_{sf}h_{fg}Pr)]^3 = [42170/(0.013 \times 2.257\times10^6 \times 1.75)]^3 = (0.821)^3 = 0.554$

$q = 636.5 \times 399.3 \times 0.554 = 1.41\times10^5$ W/m$^2$; $h = q/\Delta T_e = 14\,100$ W/m$^2$K

## Solved Example 2 – Peak heat flux

Same fluid: $q_{max} = 0.131\,(2.257\times10^6)(0.6)^{1/2}[0.0589 \times 9.81 \times 957.3]^{1/4} = 0.131 \times 2.257\times10^6 \times 0.7746 \times 4.85 = 1.11\times10^6$ W/m$^2$. A heater of area 0.1 m$^2$ can thus transfer at most 111 kW before burnout.

## Peak heat flux – other geometries (Lienhard–Dhir)

$$q_{max} = C_{cr}\, h_{fg}\, [\,\sigma\, g\, \rho_v^{\,2}(\rho_l - \rho_v)\,]^{1/4}$$

| Heater geometry | $C_{cr}$ |
|---|---|
| Large horizontal flat plate (Zuber 0.131) | 0.149 |
| Large horizontal cylinder | $0.12\,L^{*-0.25}$ |
| Large sphere | 0.11 |

→ Zuber's analysis : $q_{max} = (\pi/24)\, h_{fg}\rho_v[\,\sigma\, g(\rho_l-\rho_v)\,/\,\rho_v^{\,2}\,]^{1/4} = 0.131\, h_{fg}\rho_v^{\,1/2}[\sigma\, g(\rho_l-\rho_v)]^{1/4}$. Derived by the Helmholtz instability of vapour jets leaving the surface

## Effect of pressure & other factors on pool boiling

| p (bar) | $T_{sat}$ (°C) | $\rho_l$ | $\rho_v$ | $h_{fg}$ (kJ/kg) | $\sigma$ (N/m) |
|---|---|---|---|---|---|
| 1.013 | 100 | 958 | 0.60 | 2257 | 0.0589 |
| 5 | 152 | 915 | 2.67 | 2108 | 0.049 |
| 10 | 180 | 887 | 5.15 | 2015 | 0.042 |

→ Increase of pressure → $T_{sat}$ higher, $\sigma$ lower, $\rho_v$ higher → bubbles smaller and more numerous → $h$ increases; $q_{max}$ peaks near $p \approx 1/3\ p_{crit}$ then decreases to zero at critical pressure

→ Sub-cooling of the pool increases $q_{max}$ ; surface roughness shifts the curve to the left (lower $\Delta T_e$) ; gravity ↓ → bubbles stick, $q_{max} \propto g^{1/4}$

→ Enhancement : porous / sintered coatings, re-entrant cavities, finned tubes (Thermoexcel), ultrasonic agitation, electric fields

## Solved Example 5 – Film boiling on a horizontal tube

Horizontal tube $D = 5$ mm at 300°C in water at 1 atm ($T_{sat} = 100$°C), $\varepsilon = 0.8$. Vapour at $T_f = 200$°C : $\rho_v = 0.46$, $k_v = 0.0343$, $\mu_v = 1.62\times10^{-5}$, $c_{pv} = 1975$; $\rho_l = 958$, $h_{fg} = 2257$ kJ/kg.

$h'_{fg} = 2257\times10^3 + 0.4(1975)(200) = 2.415\times10^6$ J/kg; $\Delta T_e = 200$°C

$Nu = 0.62[9.81(0.46)(957.5)(2.415\times10^6)(0.005)^3/(1.62\times10^{-5} \times 0.0343 \times 200)]^{1/4} = 0.62(58.5) = 36.3$; $h_{conv} = 36.3(0.0343)/0.005 = 249$ W/m²K

$h_{rad} = \varepsilon\sigma(T_s^4 - T_{sat}^4)/(T_s - T_{sat}) = 0.8(5.67\times10^{-8})(673^4 - 373^4)/200 = 42$ W/m²K; $h \approx h_{conv} + 0.75\, h_{rad} = 249 + 32 = 281$ W/m²K; $q = 281(200) = 56$ kW/m²

Compare: Film boiling h (~280) is 50 times lower than nucleate boiling h (~14 000) in Example 1 – so, to transfer the same flux in the film regime, the surface has to be extremely hot!

## Solved Example 2B – Wall temperature for a given heat flux

Water at 1 atm boils on a ground stainless-steel heater ($C_{sf} = 0.008$, $n = 1$) with $q = 2\times10^5$ W/m². Other data as in Example 1. Find $\Delta T_e$ and h.

$\Delta T_e = (C_{sf} h_{fg} Pr_l/c_{pl})\,[\, q/(\mu_l h_{fg})\sqrt{(\sigma/g(\rho_l-\rho_v))}\,]^{1/3} = (0.008 \times 2.257\times10^6 \times 1.75/4217)[\, 2\times10^5/(636.5 \times 399.3)\,]^{1/3}$

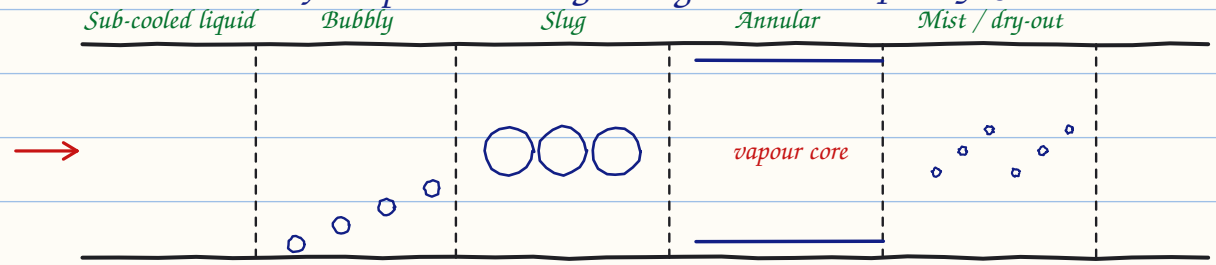
$= 7.49 \times (0.787)^{1/3} = 7.49 \times 0.923 = 6.9$°C → $T_s = 106.9$°C ; $h = q/\Delta T_e = 28\,900$ W/m²K. q is below $q_{max} = 1.1$ MW/m² so safe.

## Design steps for a boiling problem

1. Fix the regime from $\Delta T_e$ or q (compare with $q_{max}$, $q_{min}$).
2. Take saturated-liquid (or vapour at film temperature) properties at the given pressure.
3. Use Rohsenow (nucleate) / Zuber (peak) / Bromley (film) as required.
4. Check the safety margin $q/q_{max} \le 0.5$–$0.7$ for heat-flux-controlled equipment.

## Flow (forced-convection) boiling

Boiling in a tube with liquid flowing through it – used in boilers, evaporators, nuclear reactors. Bubble and flow patterns change along the tube as quality $x$ increases:




→ Sub-cooled → bubbly → slug → annular → mist. Heat transfer coefficient is high in the annular region; at dry-out the wall loses liquid film and temperature jumps

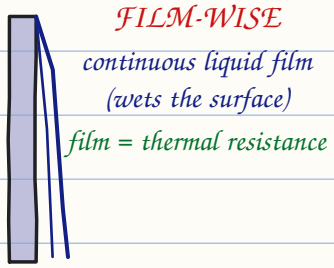
→ Chen correlation : $h = h_{nb} + h_{conv}$ (superposition of nucleate-boiling and forced-convection parts)

## Boiling vs Condensation – summary

| Aspect | Boiling | Condensation |
|---|---|---|
| Direction | liquid → vapour | vapour → liquid |
| Surface temp. | $T_s > T_{sat}$ | $T_w < T_{sat}$ |
| Main resistance | bubble/vapour film | condensate film |
| Key law | Rohsenow, Zuber, Bromley | Nusselt |
| Equipment | boilers, evaporators | condensers |
| Failure mode | burnout (CHF) | air blanketing / fouling |

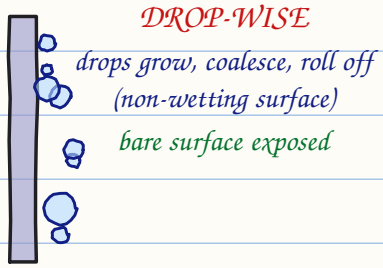
# Condensation

Condensation occurs when saturated vapour contacts a surface at a temperature $T_w$ below $T_{sat}$. Latent heat is released and the condensate forms on the surface. Two modes:



**FILM-WISE**

continuous liquid film
(wets the surface)
film = thermal resistance



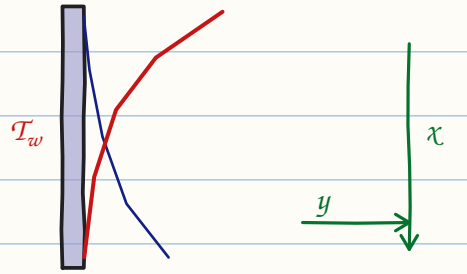
**DROP-WISE**

drops grow, coalesce, roll off
(non-wetting surface)
bare surface exposed

| Feature | Film-wise | Drop-wise |
|---|---|---|
| Surface | clean, wetted | contaminated / promoter coated |
| Mechanism | continuous film, conduction through film | drops, bare area between drops |
| h (steam) | 4000 – 11 000 $W/m^2K$ | 5 – 10 × higher (up to 250 000) |
| Occurrence | common, stable | needs promoters (oleic acid, wax, Teflon, gold plate) |
| Design basis | used (conservative) | unreliable – promoters wear off |

## Nusselt's theory of laminar film condensation on a vertical plate

Assumptions: (1) laminar film, steady; (2) constant fluid properties; (3) vapour is stagnant, exerts no shear on film ($\tau = 0$ at the outer surface); (4) heat transfer through the film is by pure conduction → linear temperature profile; (5) inertia & convection in film neglected; (6) $T_w$ constant, vapour at $T_{sat}$.




film thickness $\delta(x)$ grows with $x$

$\delta = 0$ at top, $\delta_{max}$ at bottom

$h_x = k_l/\delta(x)$ → $h$ is max at top

Gravity drains film; $T$ linear: $T_w \to T_{sat}$

Force balance on film element: (gravity – buoyancy) = viscous shear : $\mu\, d^2u/dy^2 = -g(\rho_l - \rho_v)$. With $u(0) = 0$, $du/dy|_\delta = 0$ :

$$u(y) = [\, g(\rho_l - \rho_v)/\mu_l \,]\,(\delta\, y - y^2/2)$$

Condensate flow rate per unit width : $\Gamma = \int_0^\delta \rho_l\, u\, dy = \rho_l\, g(\rho_l - \rho_v)\,\delta^3/(3\mu_l)$.

Energy balance on $dx$: latent heat released = heat conducted through film:

$$h'_{fg}\, d\Gamma = k_l (T_{sat} - T_w)\, dx/\delta$$

$$\to \delta(x) = [\, 4\,\mu_l\, k_l (T_{sat} - T_w)\, x / (\, g\, \rho_l(\rho_l - \rho_v)\, h'_{fg}\,)\,]^{1/4}$$

Steps of the derivation (energy balance on the film): $\Gamma = \rho_l\, g(\rho_l - \rho_v)\,\delta^3/3\mu_l \to d\Gamma = \rho_l\, g(\rho_l - \rho_v)\,\delta^2\, d\delta/\mu_l$. Equate $h'_{fg}\, d\Gamma = k_l(T_{sat} - T_w)\, dx/\delta$ :

$$\delta^3\, d\delta = [\, \mu_l\, k_l\, \Delta T/(\, h'_{fg}\, \rho_l\, g\, (\rho_l - \rho_v)\,)\,]\, dx \to \delta^4/4 = [\, \ldots\,]\, x$$

Local $h_x = k_l/\delta$ (conduction through film). Averaging over height $L$ :

$$h_{avg} = 0.943\,[\, \rho_l(\rho_l - \rho_v)\, g\, h'_{fg}\, k_l^3/(\, \mu_l\, L\, (T_{sat} - T_w)\,)\,]^{1/4} = (4/3)\, h_{x=L}$$

$$h'_{fg} = h_{fg} + 0.68\, c_{pl}(T_{sat} - T_w) \text{ (corrects for sub-cooling of film)} \quad \text{Nusselt}$$

→ Liquid properties at film temperature $T_f = (T_{sat} + T_w)/2$ ; $h_{fg}$ at $T_{sat}$

→ Inclined plate (angle $\varphi$ with horizontal): replace $g$ by $g \sin\varphi$. Horizontal plate facing up is not applicable

→ Condensate rate : $\dot{m} = Q / h'_{fg} = h_{avg} A (T_{sat} - T_w) / h'_{fg}$ , film Reynolds number $Re_f = 4\Gamma/\mu_l = 4\dot{m}/(b\,\mu_l)$

| $Re_f$ | Flow in film | Correction |
|---|---|---|
| < 30 | Laminar, smooth | Nusselt equation as is |
| 30 – 1800 | Laminar, wavy | $h \approx 1.2 \times$ (Nusselt) (Kutateladze: $h = 1.08\,Re^{1.22}$...) |
| > 1800 | Turbulent film | $h$ increases with $Re$ (Kirkbride): $Nu = 0.0077\,Re^{0.4}$ |

## Horizontal tube and tube banks

$$h_{avg} = 0.725\,[\,\rho_l(\rho_l - \rho_v)\,g\,h'_{fg}\,k_l^3 / (\,\mu_l\,D\,(T_{sat} - T_w)\,)\,]^{1/4}$$

Vertical row of $N$ tubes : $h_N = h_1\,N^{1/4}$ (condensate drips on lower tubes)

→ Horizontal tube is far better than a vertical tube of equal length because $D \ll L$ (thinner film). Condensers therefore use horizontal tubes

→ Sphere : coefficient 0.815 in place of 0.725 ($D$ = diameter)

## Solved Example 3 – Steam on a vertical plate

Dry saturated steam at 100°C condenses on a vertical plate 1 m high, 1 m wide at 80°C. At $T_f = 90°C$ : $\rho_l = 965$, $k_l = 0.675$, $\mu_l = 315\times10^{-6}$, $c_{pl} = 4206$; $h_{fg} = 2257$ kJ/kg. Find $h$, $Q$ and condensate rate.

$\Delta T = 20°C$; $h'_{fg} = 2257\times10^3 + 0.68(4206)(20) = 2.314\times10^6$ J/kg

$h = 0.943[965(964.4)(9.81)(2.314\times10^6)(0.675)^3/(315\times10^{-6} \times 1 \times 20)]^{1/4} = 0.943\,(5668)$

$= 5350$ W/m²K

$Q = hA\Delta T = 5350(1)(20) = 107$ kW; $\dot{m} = Q/h'_{fg} = 107\times10^3/2.314\times10^6 = 0.046$ kg/s;

$Re_f = 4\dot{m}/\mu = 4(0.046)/315\times10^{-6} = 585$ (wavy) → $h \approx 1.2 \times 5350 \approx 6400$

## Solved Example 4 – Steam on a horizontal tube

Same steam, tube $D = 20$ mm, length 1 m, $T_w = 80°C$. Same properties :

$h = 0.725[965(964.4)(9.81)(2.314\times10^6)(0.675)^3/(315\times10^{-6} \times 0.02 \times 20)]^{1/4} = 0.725\,(15\,070) = 10\,930$ W/m²K (≈ 2× the vertical plate)

$Q = h\,\pi\,D\,L\,\Delta T = 10\,930\,\pi\,(0.02)(1)(20) = 13.7$ kW; $\dot{m} = 13\,700/2.314\times10^6 = 5.9\times10^{-3}$ kg/s ≈ 21 kg/h

## Solved Example 4B – Condenser tube bank & film thickness

(a) Ten such tubes ($D$ = 20 mm, $L$ = 1 m) in one vertical row: $h_N = h_1 N^{1/4} = 10\,930 \times 10^{-0.25}$ = 6150 W/m²K.

$Q = N h_N \pi D L \Delta T = 10(6150)(\pi \times 0.02 \times 1 \times 20)$ = 77 kW; $\dot{m}$ = $77\times10^3/2.314\times10^6$ = 0.033 kg/s ≈ 120 kg/h

(b) Film thickness at the bottom of the vertical plate in Example 3 : $h_{x=L} = (3/4)h_{avg}$ = 0.75 (5350) = 4010 → $\delta_L = k_l/h_x$ = 0.675/4010 = 0.17 mm (very thin, so conduction assumption good).

## Drop-wise condensation – more detail

- Drops nucleate at surface pits, grow, merge, and roll off when large (≈ 1 mm), sweeping the surface clean so fresh area is exposed → very thin condensate layer → huge h
- Promoters : oleic acid, stearic acid, waxes, silicones, thin gold/PTFE coating. They wear off in service, hence film-wise values are used in design with a safety margin
- Empirical (steam, 1 atm) : $h_{dwc} \approx 51\,104 + 2044\, T_{sat}(°C)$ W/m²K for $T_{sat}$ 22 – 100°C (Griffith) – i.e. ≈ $2.5\times10^5$ W/m²K at 100°C

## Condensers – types & design notes

- Shell-and-tube surface condenser : vapour on shell side, cooling water through tubes. Horizontal tubes, baffles, air-vent at top.
- Direct-contact (jet) condenser : vapour mixes with cooling water spray; cheap but condensate is contaminated.
- Air-cooled condenser : finned tubes with fans; used where water is scarce.
- Condensation inside tubes : vapour shear helps the film (h higher); regimes: annular, wavy, slug, plug as quality falls.
- Overall heat transfer coefficient of condenser is usually controlled by the cooling-water side (and fouling), not by the condensing side, since $h_{cond} \gg h_{water}$.

## Factors affecting condensation

→ **Non-condensable gases (air)** : they accumulate near the film, add a gas-diffusion resistance → h can fall by 50% with even 1% air. Condensers have air-extraction (vent) pumps

→ **Vapour velocity** : high velocity downward shear thins the film → raises h; upward shear opposes drainage → lowers h

→ **Superheated vapour** : q is based on $T_{sat}$, $h_{fg}$ replaced by $h_{fg} + c_{pv}(T_{vap} - T_{sat})$; effect small

→ **Surface condition / fouling** : oil films and rough surfaces; wavy flow increases h by 20% over smooth laminar

→ **Sub-cooling of condensate** : already included through the $h'_{fg}$ correction

*Application:* Heat pipe: a sealed tube with a wick – evaporation at the hot end (boiling) and condensation at the cold end move heat with a very small $\Delta T$ (effective $k \approx 10^3$ × copper). Used in laptops, satellites, solar collectors.

## Quick viva (short answers)

Q1. Why is nucleate boiling so effective? – bubble departure agitates liquid and carries latent heat.

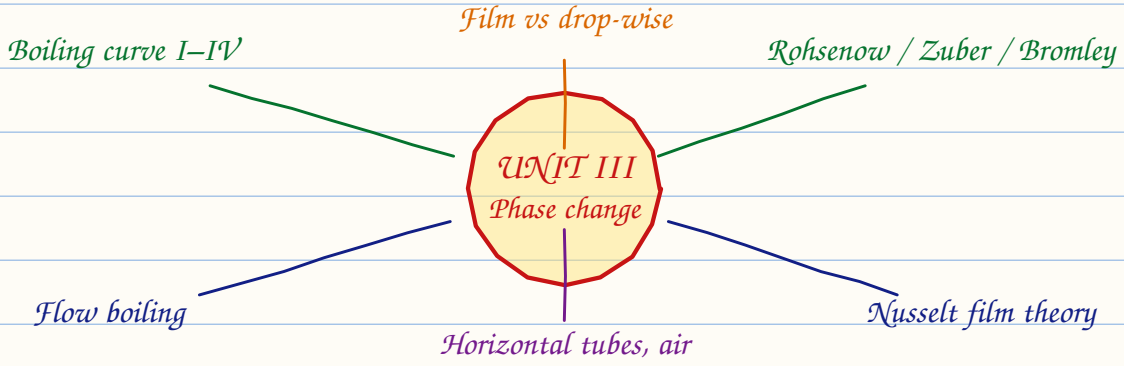
Q2. Why does q fall in the transition regime? – vapour film conductivity □ liquid conductivity.

Q3. Define Leidenfrost point – the temperature/heat flux where a stable vapour film just forms.

Q4. Does $q_{max}$ depend on the heater material? – No, only on fluid properties and geometry.

Q5. Why is $T_f$ used for properties in Nusselt's theory? – the film varies from $T_w$ to $T_{sat}$.

Q6. Effect of 1% air in steam – reduces condensation h by about 50%.



## Important Questions (exam oriented)

1. Explain the different regimes of pool boiling with the help of a boiling curve.
2. What is burnout (critical heat flux)? Why is it dangerous? Give Zuber's expression.
3. Distinguish between film-wise and drop-wise condensation. Why is drop-wise preferred but film-wise used in design?
4. Derive Nusselt's expression for the average heat transfer coefficient for laminar film condensation on a vertical plate.
5. Why are horizontal tubes preferred in condensers? Effect of non-condensable gases?
6. Explain nucleation, bubble equilibrium and flow-boiling regimes.

## Quick Revision

| Item | Expression |
|---|---|
| Excess temperature | $\Delta T_e = T_s - T_{sat}$ |
| Bubble superheat | $T_v - T_{sat} = 2\sigma T_{sat}/(\rho_v h_{fg} r)$ |
| Nucleate boiling | Rohsenow $q \propto \Delta T_e^3$ |
| Peak flux | $0.131\, h_{fg}\rho_v^{1/2}[\sigma g(\rho_l-\rho_v)]^{1/4}$ |
| Vertical film cond. | $h = 0.943[\rho(\rho-\rho_v)g\, h'_{fg}k^3/(\mu L\Delta T)]^{1/4}$ |
| Horizontal tube | $h = 0.725[\ \ldots\ /(\mu D\Delta T)\ ]^{1/4}$ |
| Film Re | $Re_f = 4\dot{m}/(b\mu_l)$ ; 30 and 1800 limits |

# UNIT IV : Heat Exchangers

Topics: types, parallel / counter / cross flow, overall U, fouling, LMTD method, ε–NTU method, solved problems

## 4.1 Introduction & Classification

A heat exchanger is a device that transfers heat between two (or more) fluids at different temperatures, usually separated by a solid wall, without mixing (e.g. radiators, condensers, boilers, evaporators, oil coolers, intercoolers).

| Basis | Classes |
|---|---|
| Heat-transfer process | Direct-contact (cooling towers, jet condenser) \| Indirect: recuperator (wall separates), regenerator (storage matrix, alternately hot/cold) |
| Flow arrangement | Parallel flow \| Counter flow \| Cross flow (mixed / unmixed) \| multi-pass, mixed |
| Construction | Double pipe, shell-and-tube, plate, plate-fin, tube-fin (compact) |
| Phase | Single-phase \| Two-phase (evaporator, condenser) |
| Compactness | Compact if $A/V > 700\ m^2/m^3$ (car radiators ≈ 1100, human lung ≈ 20 000) |

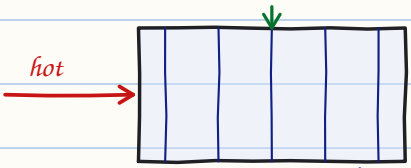
### Flow arrangements



PARALLEL FLOW (same direction)



COUNTER FLOW (opposite)




CROSS FLOW

Hot fluid : red Cold fluid : green
Cross-flow : one fluid mixed or unmixed

### Common constructions

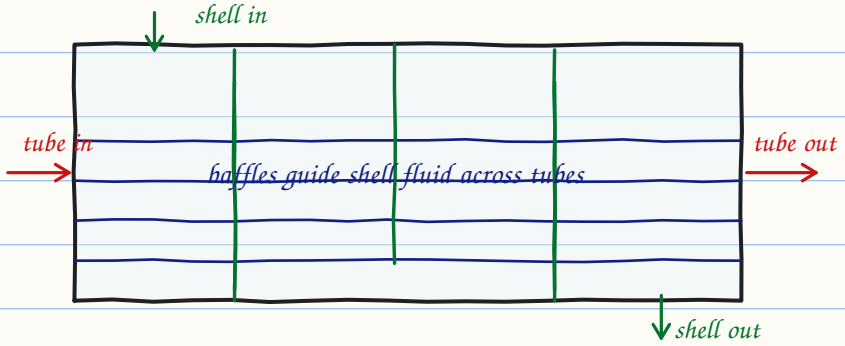
→ Double-pipe (pipe-in-pipe) : simplest; for small duties; true counter/parallel flow; hairpin sections

→ Shell-and-tube : bundle of tubes in a cylindrical shell with baffles. Designated by passes (1-1, 1-2, 2-4). TEMA types: fixed tube sheet, U-tube, floating head. Most common in industry (oil refining, power plants).

→ Plate heat exchanger : corrugated plates with gaskets; high turbulence, easy cleaning, limited to ≈ 25 bar, 150°C

→ Compact (plate-fin, tube-fin) : gas-to-gas and gas-to-liquid (radiators, intercoolers, aircraft)

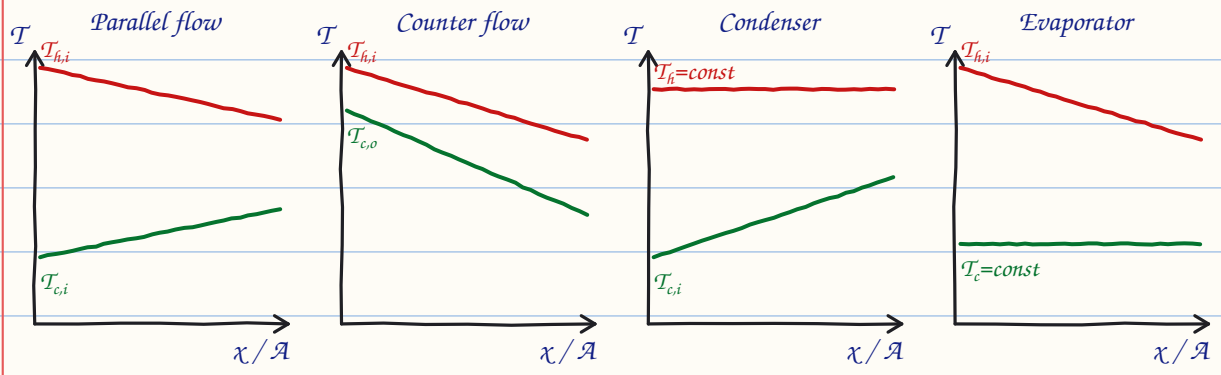
→ Regenerator : rotary wheel (air pre-heater) or fixed matrix; heat stored in matrix between hot and cold periods




1 shell pass – 2 tube passes (1-2 exchanger)

Baffles → cross-flow over tubes, higher $h_o$, $\Delta P \uparrow$

## Temperature Distributions




→ Parallel flow : both fluids enter at the same end; large $\Delta T$ at inlet, small at outlet; outlet temperatures can never cross ($T_{c,o} < T_{h,o}$).

→ Counter flow : fluids enter at opposite ends; $\Delta T$ fairly uniform; $T_{c,o}$ can exceed $T_{h,o}$; gives highest $\Delta T_{lm}$ and best effectiveness for the same $A$ and $U$.

→ Condenser (hot fluid constant $T$) and evaporator (cold fluid constant $T$): arrangement does not matter – $C_{max} \to \infty$, $C_r = 0$.

## Overall Heat Transfer Coefficient U

Heat passes through: hot-fluid film → fouling → wall → fouling → cold-fluid film. The resistances add in series. For a plane wall:

$$1/U = 1/h_i + R_{fi} + L/k + R_{fo} + 1/h_o \qquad \text{plane wall}$$

For a tube (inner radius $r_i$, outer radius $r_o$, length $L$) the area must be specified:

$$1/(U_o A_o) = 1/(h_i A_i) + R_{fi}/A_i + \ln(r_o/r_i)/(2\pi k L) + R_{fo}/A_o + 1/(h_o A_o)$$

$$Q = U_o\, A_o\, \Delta T_m = U_i\, A_i\, \Delta T_m \;(U_i A_i = U_o A_o = UA)$$ *tube*

→ When the wall is thin and k high: $1/U \approx 1/h_i + 1/h_o$ → U is dominated by the smaller h (gas side). Remedy: fins on the gas side.

| Fluid combination | U (W/m²K) |
|---|---|
| Water – water | 850 – 1700 |
| Water – oil | 100 – 350 |
| Steam condenser (water cooled) | 1000 – 6000 |
| Steam – light fuel oil | 200 – 400 |
| Ammonia condenser | 800 – 1400 |
| Air – water (finned) | 25 – 60 |
| Gas – gas | 10 – 40 |

## Solved Example 0 – Overall U of a double-pipe exchanger

Inner tube $D_i$ = 25 mm, $D_o$ = 30 mm, steel $k$ = 50 W/mK; $h_i$ = 1500, $h_o$ = 800 W/m²K; fouling $R_{fi}$ = 0.0002, $R_{fo}$ = 0.0001 m²K/W. Find $U_o$ and the length needed for $Q$ = 63 kW if $\Delta T_{lm}$ = 83.9°C.

$1/U_o = (r_o/r_i)(1/h_i + R_{fi}) + r_o \ln(r_o/r_i)/k + R_{fo} + 1/h_o$

$= 1.2(0.000667 + 0.0002) + 0.015(0.1823)/50 + 0.0001 + 0.00125 = 0.00104 +$

$0.0000547 + 0.0001 + 0.00125 = 0.002445 \rightarrow U_o = 409$ W/m²K

$A_o = Q/(U_o \Delta T_{lm}) = 63\,000/(409 \times 83.9) = 1.84$ m²; $L = A_o/(\pi D_o) = 1.84/(\pi \times 0.03) =$ 19.5 m (i.e. several hairpin sections)

# Fouling in Heat Exchangers

Fouling = deposit of unwanted material (scale, sludge, corrosion products, biological growth) on the heat-transfer surface. It adds a thermal resistance $R_f$ (m²K/W) and also restricts the flow (higher ΔP).

$$1/U_{dirty} = 1/U_{clean} + R_{fi} + R_{fo}\,,\; R_f = (1/U_d) - (1/U_c)$$

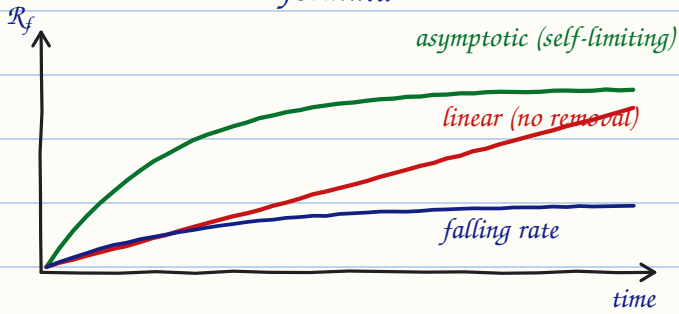
| Fluid | $R_f$ ($m^2K/W$) |
|---|---|
| Sea water / treated boiler feed water (< 50°C) | 0.0001 |
| Sea water (> 50°C) | 0.0002 |
| River / city water | 0.0002 – 0.0004 |
| Fuel oil | 0.0009 |
| Steam (oil-free) | 0.0001 |
| Refrigerant liquid | 0.0002 |

→ Types : precipitation (scaling of hard-water salts), particulate, corrosion, biological (algae, bacteria), chemical reaction (polymerisation, coking), freezing

→ Effects : lower U, bigger area needed, higher pumping power, shutdowns, over-design (10–30 %)

→ Control : higher velocity (> 1 m/s), treated water, filters, anti-scalants, tube cleaning (brushes, ball cleaning), choose smooth materials, avoid wall temperature hot-spots, keep dead zones small

→ Cleanliness factor $CF = U_d/U_c$ (0.8 – 0.95 typical); fouling factor adds to resistance in series as in the U formula




Induction → growth → saturation

Net deposition = deposit rate – removal rate (shear)

Higher velocity → higher removal

## LMTD Method

Basis : $Q = U A \Delta T_m$, where $\Delta T_m$ is a suitable mean of the varying temperature difference between hot and cold streams. Assumptions: steady flow; constant U and $c_p$; no heat loss to surroundings; no axial conduction; no phase change except in condenser/evaporator; each fluid has uniform properties across section.

## Derivation (parallel flow)

Element $dA$: $dQ = -C_h\,dT_h = C_c\,dT_c$ ($C = \dot{m}\,c_p$, heat-capacity rate), $dQ = U\,\Delta T\,dA$ with $\Delta T = T_h - T_c$:

$$d(\Delta T) = dT_h - dT_c = -dQ\,(1/C_h + 1/C_c) = -U\,\Delta T\,dA\,(1/C_h + 1/C_c)$$

Integrate from inlet (1) to outlet (2): $\ln(\Delta T_2/\Delta T_1) = -UA\,(1/C_h + 1/C_c)$. But $Q = C_h(T_{h1} - T_{h2}) = C_c(T_{c2} - T_{c1})$ so $1/C_h + 1/C_c = [(T_{h1}-T_{h2}) + (T_{c2}-T_{c1})]/Q = (\Delta T_1 - \Delta T_2)/Q$. Hence:

$$Q = U\,A\,\Delta T_{lm}$$

$$\Delta T_{lm} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2) \qquad \text{LMTD}$$

| Arrangement | $\Delta T_1$ (end 1) | $\Delta T_2$ (end 2) |
|---|---|---|
| Parallel flow | $T_{h,i} - T_{c,i}$ | $T_{h,o} - T_{c,o}$ |
| Counter flow | $T_{h,i} - T_{c,o}$ | $T_{h,o} - T_{c,i}$ |

→ If $\Delta T_1 = \Delta T_2$, $\Delta T_{lm} = \Delta T_1$ (limit). If $\Delta T_1/\Delta T_2 < 1.7$, the arithmetic mean $(\Delta T_1+\Delta T_2)/2$ is within 2 % of LMTD.

→ Counter-flow $\Delta T_{lm}$ > parallel-flow $\Delta T_{lm}$ for the same terminal temperatures → smaller area. Condenser/evaporator: same $\Delta T_{lm}$ for either arrangement.

→ Counter-flow, $C_h = C_c$ : both temperature curves are parallel, $\Delta T$ is constant = $\Delta T_1 = \Delta T_2$ and $\Delta T_{lm} = \Delta T_1$.

→ Heat capacity rate $C = \dot{m}\,c_p$ (W/K) : the fluid with smaller $C$ undergoes the larger temperature change. Also: $C_h/C_c = (T_{c,o} - T_{c,i})/(T_{h,i} - T_{h,o})$.

## Cross-flow and multi-pass exchangers

Use the counter-flow LMTD multiplied by a correction factor $F$ ($0 < F \le 1$) read from charts for given $P$ and $R$:

$$Q = U\,A\,F\,\Delta T_{lm,CF}$$

$$P = (t_2 - t_1)/(T_1 - t_1)\,,\quad R = (T_1 - T_2)/(t_2 - t_1) = C_c/C_h$$

For 1 shell pass, 2 (or 2n) tube passes the analytical value is:

$$F = [\sqrt{(R^2+1)}/(R-1)]\,\ln[(1-P)/(1-PR)] \,/\, \ln\{[2 - P(R+1-\sqrt{(R^2+1)})] / [2 - P(R+1+\sqrt{(R^2+1)})]\}$$

→ $F = 1$ for a pure counter-flow, $F = 1$ when one fluid isothermal ($R \to 0$ or $\infty$). Good design keeps $F > 0.75$; if $F < 0.75$ use more shells in series.

## Solved Example 1 – Counter-flow vs parallel flow (area)

Oil ($\dot{m}$ = 0.5 kg/s, $c_p$ = 2100) is cooled from 150°C to 90°C by water ($\dot{m}$ = 0.8 kg/s, $c_p$ = 4180) entering at 25°C. $U$ = 300 W/m²K. Find the area for counter and parallel flow.

$Q$ = 0.5(2100)(60) = 63 kW; water rise = 63 000/(0.8 × 4180) = 18.8°C → $T_{c,o}$ = 43.8°C

Counter : $\Delta T_1$ = 150 - 43.8 = 106.2; $\Delta T_2$ = 90 - 25 = 65; $\Delta T_{lm}$ = 41.2/ln(1.634) = 83.9°C; $A$ = 63 000/(300 × 83.9) = 2.50 m²

Parallel : $\Delta T_1$ = 150 - 25 = 125; $\Delta T_2$ = 90 - 43.8 = 46.2; $\Delta T_{lm}$ = 78.8/ln(2.705) = 79.1°C; $A$ = 63 000/(300 × 79.1) = 2.65 m² (6 % more)

## Solved Example 2 – Shell & tube (1-2) with F

Hot water 120°C → 60°C ($C_h$ = 5 kW/K), cold stream 20°C → 50°C, $U$ = 1000 W/m²K. 1 shell – 2 tube passes. Find $A$.

$Q$ = 5(60) = 300 kW; $\Delta T_1$ = 120 - 50 = 70, $\Delta T_2$ = 60 - 20 = 40 → $\Delta T_{lm,CF}$ = 30/ln(1.75) = 53.6°C

$P$ = 30/100 = 0.3, $R$ = 60/30 = 2.0 → $F$ = (2.236/1) ln(0.7/0.4)/ln(1.771/0.429) = 1.251/1.417 = 0.883

$A$ = $Q/(U F \Delta T_{lm})$ = 300 000/(1000 × 0.883 × 53.6) = 6.34 m²

## Cross-check of Example 2 using the NTU method

$C_h$ = 5 kW/K. $C_c$ = $Q/(\Delta t_c)$ = 300/30 = 10 kW/K → $C_{min}$ = 5, $C_r$ = 0.5. $\varepsilon$ = $Q/[C_{min}(T_{h,i} - T_{c,i})]$ = 300/(5 × 100) = 0.60

Counter-flow NTU = $[1/(C_r-1)]$ ln$[(\varepsilon-1)/(\varepsilon C_r-1)]$ = (1/-0.5) ln(0.4/0.7) = 1.119 → $UA$ = 1.119 × 5000 = 5596 W/K. The 1-2 shell exchanger needs $UA$ = 1000 × 6.34 = 6340 → ratio 5596/6340 = 0.883 = $F$ (OK). (Shows $F$ physically = loss of performance versus pure counter-flow).

# Effectiveness – NTU Method

The LMTD method needs all four terminal temperatures. When only inlet temperatures are known (rating / performance problems) use the $\varepsilon$–NTU method (Kays & London).

## Definitions

$$C_h = \dot{m}_h c_{ph}\ ,\ C_c = \dot{m}_c c_{pc}\ ;\ C_{min},\ C_{max}\ ;\ C_r = C_{min}/C_{max}\ (0 \le C_r \le 1)$$

$$Q_{max} = C_{min}\,(T_{h,i} - T_{c,i})\ \varepsilon = Q/Q_{max} \rightarrow Q = \varepsilon\ C_{min}\,(T_{h,i} - T_{c,i})$$

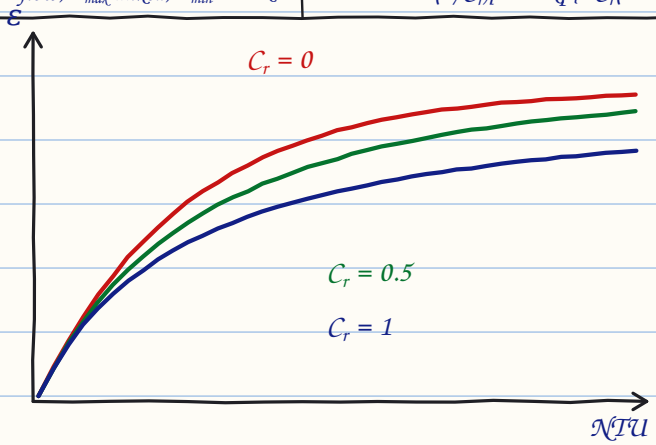
$$NTU = UA / C_{min}$$

→ $\varepsilon$ = actual heat transfer / maximum possible ($0 < \varepsilon < 1$). Fluid with $C_{min}$ undergoes the larger temperature change; $Q_{max}$ would need infinite area counter-flow.

→ $NTU$ (number of transfer units) = size of the exchanger in terms of heat-capacity rate: a larger $NTU$ → higher $\varepsilon$ with diminishing returns beyond $NTU \approx 3$.

→ $\varepsilon = f(NTU, C_r,$ flow arrangement) :

| Flow arrangement | $\varepsilon = f(NTU, C_r)$ | $NTU = f(\varepsilon, C_r)$ |
|---|---|---|
| Parallel flow | $[1 - e^{-NTU(1+C_r)}] / (1+C_r)$ | $-\ln[1 - \varepsilon(1+C_r)] / (1+C_r)$ |
| Counter flow | $[1 - e^{-NTU(1-C_r)}] / [1 - C_r e^{-NTU(1-C_r)}]$ | $1/(C_r-1) \cdot \ln[(\varepsilon-1)/(\varepsilon C_r-1)]$ |
| Counter, $C_r = 1$ | $NTU / (1 + NTU)$ | $\varepsilon / (1 - \varepsilon)$ |
| Any, $C_r = 0$ (boiler/condenser) | $1 - e^{-NTU}$ | $-\ln(1 - \varepsilon)$ |
| Shell-tube 1 shell pass (2,4.. tube passes) | $2\{1 + C_r + \sqrt{(1+C_r^2)} \cdot (1+e^{-\Gamma})/(1-e^{-\Gamma})\}^{-1}$, $\Gamma = NTU\sqrt{(1+C_r^2)}$ | — |
| Cross flow, both unmixed (approx.) | $1 - \exp\{NTU^{0.22}[e^{-C_r NTU^{0.78}} - 1] / C_r\}$ | — |
| Cross flow, $C_{max}$ mixed, $C_{min}$ unmixed | $(1/C_r)[1 - \exp\{-C_r(1 - e^{-NTU})\}]$ | — |




Counter-flow exchanger

- $\varepsilon$ rises steeply for $NTU < 2$
- little gain beyond $NTU \approx 3$–4
- lower $C_r$ → higher $\varepsilon$
- $\varepsilon = 1$ only if $NTU \rightarrow \infty$ , $C_r<1$

Remember: Parallel flow : the maximum possible $\varepsilon = 1/(1 + C_r)$ ($NTU \rightarrow \infty$), e.g. only 0.5 when $C_r = 1$; counter-flow can reach 1. This is why counter-flow is superior.

## Derivation of ε for parallel flow

From the LMTD derivation: $\ln(\Delta T_2/\Delta T_1) = -UA(1/C_h + 1/C_c)$. Let $C_{min} = C_h$ (say) so $C_r = C_h/C_c$: $(T_{h,o} - T_{c,o})/(T_{h,i} - T_{c,i}) = \exp[-NTU(1 + C_r)]$.

Energy balance : $T_{c,o} - T_{c,i} = C_r(T_{h,i} - T_{h,o})$ and $T_{h,o} - T_{c,o} = (T_{h,i} - T_{c,i}) - (1 + C_r)(T_{h,i} - T_{h,o})$.

Substituting, with $\varepsilon = (T_{h,i} - T_{h,o})/(T_{h,i} - T_{c,i})$ :

$$1 - \varepsilon(1 + C_r) = \exp[-NTU(1+C_r)] \Rightarrow \varepsilon = \{1 - \exp[-NTU(1+C_r)]\} / (1 + C_r)$$ parallel flow

Counter-flow : repeat with $\Delta T_1 = T_{h,i} - T_{c,o}$, $\Delta T_2 = T_{h,o} - T_{c,i} \rightarrow \ln(\Delta T_2/\Delta T_1)$... giving $\varepsilon = (1 - e^{-NTU(1-C_r)}\})/(1 - C_r e^{-NTU(1-C_r)}\})$.

## Mixed and unmixed cross-flow

→ Unmixed : fins / channels stop the fluid moving transversely, so temperature varies in the direction normal to the flow. Mixed : fluid free to mix (a bare-tube bank) → uniform temperature across the stream at any section.

→ Both unmixed (plate-fin core) > one mixed > both mixed (for the same NTU); the stream with $C_{max}$ mixed / $C_{min}$ unmixed is better than the opposite.

## Procedure – NTU method (rating problem)

1. Compute $C_h$, $C_c \rightarrow C_{min}$, $C_{max}$, $C_r$. 2. $NTU = UA/C_{min}$. 3. Read ε from the formula/chart for the arrangement.

4. $Q = \varepsilon\, C_{min}(T_{h,i} - T_{c,i})$. 5. $T_{h,o} = T_{h,i} - Q/C_h$ ; $T_{c,o} = T_{c,i} + Q/C_c$.

## Solved Example 3 – Counter-flow, NTU method

Hot oil $C_h = 2400$ W/K at 160°C; cold water $C_c = 4000$ W/K at 30°C; counter-flow with $UA = 4200$ W/K. Find Q and outlet temperatures.

$C_{min} = 2400$, $C_{max} = 4000$, $C_r = 0.6$; $NTU = 4200/2400 = 1.75$

$\varepsilon = [1 - e^{-1.75(0.4)}]/[1 - 0.6\, e^{-1.75(0.4)}] = (1 - 0.4966)/(1 - 0.2979) = 0.717$

$Q = 0.717(2400)(160 - 30) = 223.7$ kW; $T_{h,o} = 160 - 223\,700/2400 = 66.8$°C; $T_{c,o} = 30 + 223\,700/4000 = 85.9$°C

(For parallel flow $\varepsilon = [1 - e^{-1.75(1.6)}]/1.6 = 0.587 \rightarrow Q = 183$ kW, i.e. 18 % less.)

## Solved Example 4 – Condenser ($C_r = 0$)

Steam condenses at 100°C on tubes carrying water ($\dot{m}$ = 2 kg/s, $c_p$ = 4180 → $C$ = 8360 W/K) entering at 20°C. $U$ = 2500 W/m²K, $A$ = 5 m².

$NTU = 2500(5)/8360 = 1.495$; $\varepsilon = 1 - e^{-1.495} = 0.776$; $Q = 0.776(8360)(100 - 20) =$ 519 kW; $T_{c,o} = 20 + 0.776(80) = 82.1°C$

## Solved Example 5 – Design with fouling

A clean exchanger has $U_c$ = 850 W/m²K. After service, fouling resistances $R_{fi}$ = 0.0002 and $R_{fo}$ = 0.0001 m²K/W develop.

$1/U_d = 1/850 + 0.0002 + 0.0001 = 0.001176 + 0.0003 = 0.001476$ → $U_d$ = 677 W/m²K; $CF = 677/850 = 0.80$ → area must be 25 % larger (850/677 = 1.255) to give the same duty.

## LMTD vs NTU – when to use which

| Given | Find | Method |
|---|---|---|
| 4 terminal temperatures (or Q and 3 T) | Area A (sizing / design) | LMTD |
| Inlet temps, flow rates, A, U | Outlet temps, Q (rating) | ε–NTU |
| Condenser / evaporator | Either | either; $C_r = 0$ simplifies NTU |

## Heat exchanger selection – design considerations

- Required duty and temperatures; fluid properties; allowable pressure drop; fouling tendency; corrosion → material (SS, Cu-Ni, Ti)
- Size, weight, cost, maintenance (removable bundles for cleaning), leakage/cross-contamination risk, thermal stresses (floating head or U-tube)
- Put the fouling / corrosive / high-pressure fluid in the tubes (cheaper to clean and pressurise), the viscous or condensing fluid in the shell

## Compact heat exchangers – key points

- Plate-fin / tube-fin : area density $\beta = A/V$ up to 6000 m²/m³; used for gas-side with low h. Fin efficiency $\eta_o = 1 - (A_f/A)(1 - \eta_f)$ must be included in U.
- Air-conditioning coils, radiators, aircraft oil coolers, cryogenic exchangers (Re small, $\Delta P$ small).

## Solved Example 6 – Steam condenser (LMTD)

Steam condenses at 120°C; cooling water ($\dot{m}$ = 3 kg/s, $c_p$ = 4180) heats from 20°C to 60°C; $U$ = 2000 W/m²K. Find $A$.

$Q = 3(4180)(40) = 501.6$ kW; $\Delta T_1 = 120 - 20 = 100$, $\Delta T_2 = 120 - 60 = 60$; $\Delta T_{lm} = 40/\ln(1.667) = 78.3$°C

$A = 501\,600/(2000 \times 78.3) = 3.20$ m²

## Solved Example 7 – Cross-flow, both fluids unmixed (NTU)

Hot gas $C_h$ = 1000 W/K at 400°C; air $C_c$ = 2500 W/K at 30°C; $UA$ = 1500 W/K. $C_{min}$ = 1000, $C_r$ = 0.4, $NTU$ = 1.5.

$\varepsilon = 1 - \exp\{ NTU^{0.22}[\, e^{-C_r NTU^{0.78}} - 1 \,]/C_r \} = 1 - \exp\{ 1.093(0.578 - 1)/0.4 \} = 1 - e^{-1.154} = 0.685$

$Q = 0.685(1000)(370) = 253$ kW; $T_{h,o} = 400 - 253 = 147$°C; $T_{c,o} = 30 + 253\,300/2500 = 131$°C

## Solved Example 8 – Unknown outlet temperatures

Counter-flow water-to-water: hot 0.5 kg/s at 90°C, cold 0.25 kg/s at 20°C, $U$ = 1000, $A$ = 2 m². ($c_p$ = 4180). $C_h = 2090$, $C_c = 1045 \rightarrow C_{min} = 1045$, $C_r = 0.5$, $NTU = 2000/1045 = 1.914$.

$\varepsilon = (1 - e^{-0.957})/(1 - 0.5\, e^{-0.957}) = 0.616/0.808 = 0.762$; $Q = 0.762(1045)(70) = 55.8$ kW; $T_{h,o} = 90 - 26.7 = 63.3$°C; $T_{c,o} = 20 + 53.4 = 73.4$°C

## Regenerators and multi-pass behaviour

- → *Regenerator* : hot gas heats a matrix (period 1), then cold gas takes heat from it (period 2). Used in gas-turbine recuperation, glass furnaces, rotary air pre-heaters (efficiency 70–85 %).
- → In a 1-2 shell-and-tube unit the tube-side fluid passes once with and once against the shell flow; temperature *cross-over* can occur, but a large cross-over reduces $F$ sharply ($F < 0.75$ → add shells).
- → Using several shells in series approaches true counter-flow: $\varepsilon$ for $N$ shells increases and $F$ rises towards 1.

## Thermal design procedure (flow chart in words)

1. Fix duty Q from energy balance; find missing temperature. 2. Choose type, fluid allocation, tube size & velocity.

3. Compute $h_i$, $h_o$ (Unit II/III correlations) → $U$ with fouling. 4. $\Delta T_{lm}$, $F$ → area $A = Q/(UF\Delta T_{lm})$.

5. Fix number of tubes & length; check $\Delta p$. 6. If $\Delta p$ or velocity not acceptable, change passes / baffles and repeat.

## Comparison of exchanger types

| Type | Advantages | Limitations |
|---|---|---|
| Double pipe | simple, cheap, true counter-flow | small area, bulky for large duty |
| Shell & tube | robust, high p,T, large duty | large, costly, shell-side fouling |
| Plate | compact, high U (3000–7000), easy cleaning | gaskets limit p,T (< 25 bar) |
| Plate-fin / compact | very high A/V, light weight | clean fluids only, difficult to clean |

## Pressure drop and pumping power

→ Tube side : $\Delta p = [\, f\, L/D + 4\, n_p \,]\, \rho\, u_m^2/2$ ; more tube passes → higher $h$ but higher $\Delta p \propto u^2$

→ Shell side : baffles increase $h_o$ but also $\Delta p$; baffle spacing ≈ 0.2 – 1 shell diameter, 25 % cut

→ Pumping power = $V\Delta p / \eta_{pump}$. Optimum design balances capital cost (area) with operating cost (pumping).

## Quick Revision – formulae

| Item | Formula |
|---|---|
| Overall U (plane wall) | $1/U = 1/h_i + R_{fi} + L/k + R_{fo} + 1/h_o$ |
| LMTD | $(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$ |
| Duty | $Q = UA\, F\, \Delta T_{lm}$ |
| Effectiveness | $\varepsilon = Q/[C_{min}(T_{h,i} - T_{c,i})]$ |
| NTU | $UA/C_{min}$ |
| Counter-flow ε | $(1-e^{-NTU(1-C_r)})/(1-C_r e^{-NTU(1-C_r)})$ |
| Fouling | $1/U_d = 1/U_c + R_f$ |

## Quick viva (short answers)

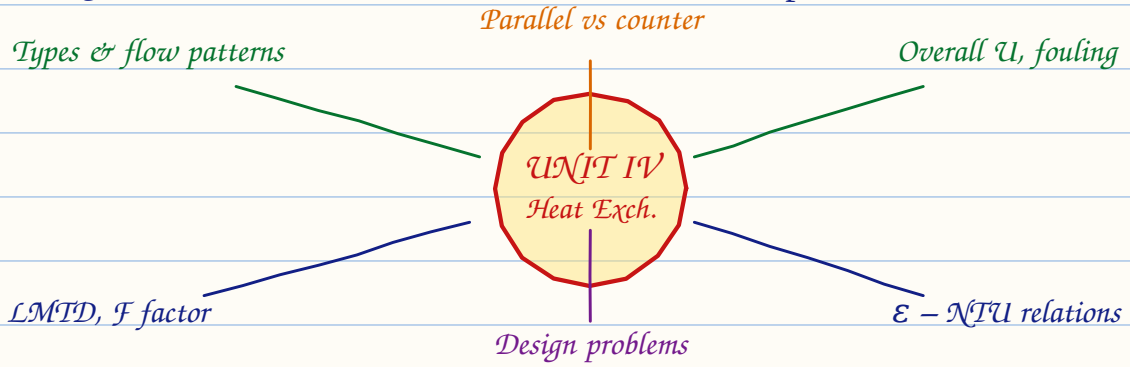
Q1.Why do outlet temperatures cross in counter-flow only? – in counter-flow cold leaves near hot inlet.

Q2.Value of $F$ for condenser? – 1. Value of $C_r$ for evaporator? – 0.

Q3.Which fluid should be on tube side? – the fouling, corrosive or high-pressure one.

## Important Questions (exam oriented)

1. Classify heat exchangers. Sketch temperature distributions in parallel and counter-flow exchangers.
2. Derive the LMTD expression for a parallel-flow heat exchanger. State the assumptions.
3. Why is counter-flow better than parallel flow? What is the correction factor $F$?
4. Define effectiveness and NTU. Derive $\varepsilon$ for parallel-flow in terms of NTU and $C_r$.
5. What is fouling? List its types and effect on overall heat transfer coefficient.
6. Distinguish LMTD and NTU methods with a worked problem each.



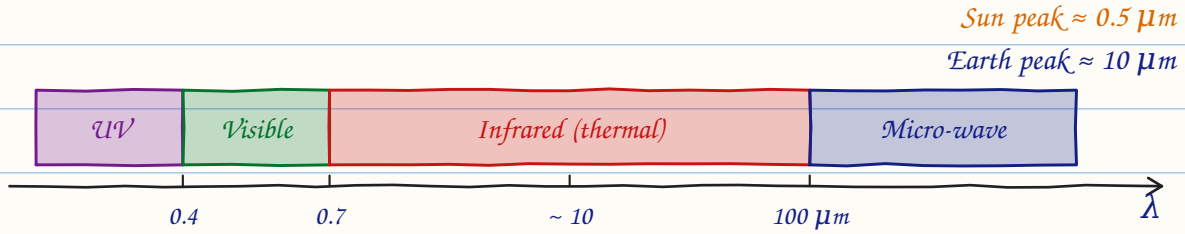
# UNIT V : Radiation & Mass Transfer

Topics: thermal radiation, laws, black & gray bodies, shape factor, radiation exchange, shields, greenhouse effect; mass transfer: Fick's law, equimolal & stagnant-gas diffusion, $h_m$

## 5.1 Thermal Radiation – Basics

Thermal radiation is energy emitted by matter because of its temperature, carried by electromagnetic waves (or photons). It needs no medium and travels at the speed of light – it is the only mode that works in vacuum (Sun → Earth). Thermal band : $\lambda$ = 0.1 – 100 $\mu$m (UV, visible 0.4–0.7 $\mu$m, infrared).

Thermal radiation 0.1 – 100 $\mu$m




When radiation (irradiation G) falls on a body it is partly absorbed ($\alpha$), reflected ($\rho$) and transmitted ($\tau$):

$$\alpha + \rho + \tau = 1$$

→ Opaque body ($\tau$ = 0) : $\alpha + \rho = 1$. Black body : $\alpha = 1$, $\rho = \tau = 0$ (absorbs all, emits maximum at every T and $\lambda$). White body : $\rho = 1$. Transparent : $\tau = 1$.

→ Reflection may be specular (mirror, angle in = angle out) or diffuse (rough surface, scattered in all directions).

→ Surface radiation: for opaque solids emission and absorption are surface phenomena (within a few $\mu$m); gases and glass are volume phenomena.

### Definitions

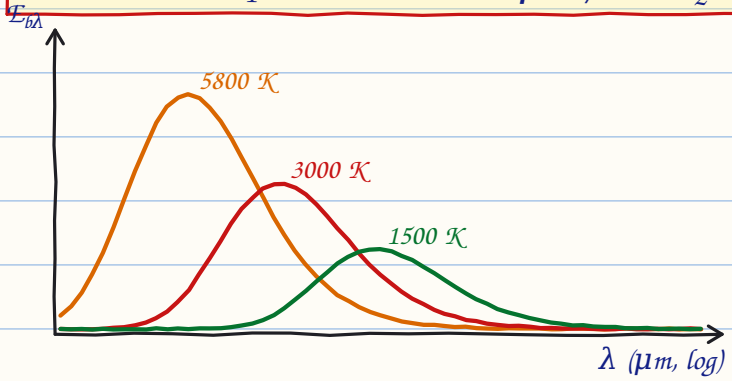
| Quantity | Symbol / Unit | Meaning |
|---|---|---|
| Emissive power | $E$ ($W/m^2$) | energy emitted per unit area, all $\lambda$, all directions |
| Monochromatic $E_\lambda$ | $W/m^2\mu m$ | emission per unit wavelength |
| Intensity | $I$ ($W/m^2 sr$) | emission per unit projected area per solid angle |
| Irradiation $G$ | $W/m^2$ | radiation incident on a surface |
| Radiosity $J$ | $W/m^2$ | total radiation leaving a surface (emitted + reflected) |
| Emissivity $\varepsilon$ | — | $E/E_b$ at same $T$ ($0 \le \varepsilon \le 1$) |

# Laws of Radiation

## (1) Planck's law – spectral distribution of a black body

$$E_{b\lambda} = C_1 \lambda^{-5} / [\exp(C_2/\lambda T) - 1] \ (W/m^2\mu m)$$

$$C_1 = 3.742\times10^8 \ W\cdot\mu m^4/m^2, \ C_2 = 14\,388 \ \mu m\cdot K$$ Planck




- Every curve has a peak (Wien)
- Peak shifts to short $\lambda$ as $T$ rises
- Area under curve = $\sigma T^4$
- Hotter curve lies above all
- (peaks not to scale)

## (2) Wien's displacement law

$$\lambda_{max} T = 2898 \ \mu m\cdot K$$ Wien

→ Sun (5800 K) : $\lambda_{max}$ = 2898/5800 = 0.50 μm (visible); tungsten filament 2800 K : 1.03 μm (mostly IR); human body 310 K : 9.3 μm

## (3) Stefan–Boltzmann law – total emission

$$E_b = \int_0^\infty E_{b\lambda}\, d\lambda = \sigma T^4 \ \sigma = 5.67\times10^{-8} \ W/m^2K^4$$

$$\text{Real surface : } E = \varepsilon \sigma T^4$$ Stefan–Boltzmann

## (4) Kirchhoff's law

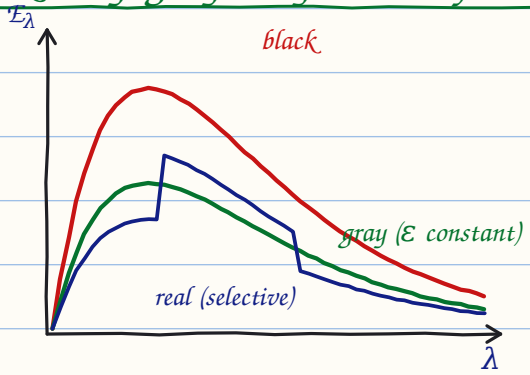
At thermal equilibrium, emissivity = absorptivity: $\varepsilon_{\lambda,\theta} = \alpha_{\lambda,\theta}$. For gray, diffuse surfaces (or when the incident radiation is from a body at the same temperature) $\varepsilon = \alpha$. A good absorber is a good emitter.

## (5) Lambert's cosine law

$$I_\theta = I_n \cos\theta \text{ (emission per unit solid angle} \propto \cos\theta)$$

$$E = \pi I \text{ (diffuse surface)}$$ Lambert

## Black body, gray body, real surface




Black : $\varepsilon = 1$

Gray : $\varepsilon < 1$, independent of $\lambda$

Real : $\varepsilon$ varies with $\lambda$, T, finish

Polished Al ≈ 0.05 ; oxidised steel ≈ 0.8

Black paint 0.95 ; human skin 0.95

Net radiation from a small gray body ($A$, $\varepsilon$, $T$) in a large surrounding at $T_{sur}$: $Q = \varepsilon \sigma A (T^4 - T_{sur}^4)$

$$h_r = \varepsilon \sigma (T + T_{sur})(T^2 + T_{sur}^2) \rightarrow Q_{rad} = h_r A (T - T_{sur})$$

## Solved Example 1 – Emission and Wien

A surface of 2 $m^2$ at 800 K, $\varepsilon = 0.8$ : $Q = 0.8(5.67\times10^{-8})(2)(800)^4 = 0.8 \times 5.67\times10^{-8} \times 2 \times 4.096\times10^{11} = 37.2$ kW. $\lambda_{max} = 2898/800 = 3.62$ μm (infrared).

## Solved Example 2 – Combined convection + radiation

Pipe $D = 0.1$ m, $T_s = 400$ K, $\varepsilon = 0.8$, in air and surroundings at 300 K, $h_c = 10$ W/$m^2$K.

$h_r = 0.8(5.67\times10^{-8})(700)(400^2+300^2) = 4.536\times10^{-8}(700)(2.5\times10^5) = 7.9$ W/$m^2$K

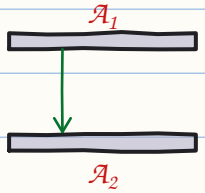
$Q/L = (h_c + h_r)\pi D (T_s - T_\infty) = (10 + 7.9)(\pi \times 0.1)(100) = 563$ W/m (radiation ≈ 44 % of the loss)

# Radiation Exchange between Surfaces

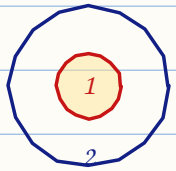
## Shape (view) factor $F_{12}$

$F_{12}$ = fraction of radiation leaving surface 1 (diffuse, isothermal) that directly strikes surface 2. It depends only on geometry (size, distance, orientation):

$$F_{12} = (1/A_1) \int_{A1}\int_{A2} ( \cos\theta_1 \cos\theta_2 / \pi r^2 ) \, dA_1 \, dA_2$$




infinite parallel plates $F_{12} = 1$




enclosed body: $F_{12} = 1$ , $F_{21} = A_1/A_2$

| Property | Relation | Remark |
|---|---|---|
| Reciprocity | $A_1F_{12} = A_2F_{21}$ | from symmetry of the integral |
| Summation | $\Sigma_j F_{ij} = 1$ | enclosure of $N$ surfaces |
| Plane / convex surface | $F_{ii} = 0$ | cannot see itself |
| Concave surface | $F_{ii} > 0$ | sees itself |
| Superposition | $F_{1(2+3)} = F_{12} + F_{13}$ | add areas seen |

→ Infinite long 3-surface duct (triangle, sides $L_1$, $L_2$, $L_3$) : $F_{12} = (L_1 + L_2 - L_3)/(2L_1)$ (Hottel crossed-string). Equilateral triangle: $F_{12} = 0.5$. For a 3-4-5 triangle, $F_{12}$ $(L_1=3, L_2=4, L_3=5) = 2/6 = 0.33$.

→ Two coaxial parallel discs, perpendicular rectangles, etc. : read from charts ($F$ as function of $X/L$, $Y/L$).

→ For a cylinder (1) inside concentric cylinder (2): $F_{12} = 1$, $F_{21} = r_1/r_2$, $F_{22} = 1 - r_1/r_2$.

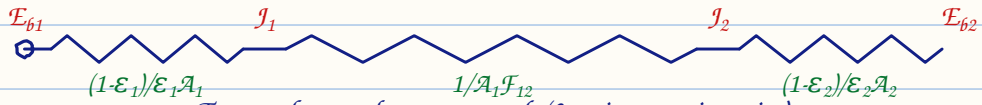
## Black surfaces

$$Q_{12} = A_1F_{12}\,\sigma\,(T_1^4 - T_2^4) = (E_{b1} - E_{b2}) / R_{12}\,,\; R_{12} = 1/(A_1F_{12})$$ space resistance

## Gray diffuse surfaces – radiosity–irradiation method (Oppenheim)

Radiosity $J = \varepsilon E_b + \rho G = \varepsilon E_b + (1 - \varepsilon) G$. Net heat leaving a surface : $Q = A (J - G)$ $= A \varepsilon (E_b - G)/\ldots$ which simplifies to:

$$Q_i = (E_{bi} - J_i) / [(1 - \varepsilon_i) / (\varepsilon_i A_i)] \text{ (surface resistance)}$$
$$Q_{ij} = (J_i - J_j) / [1 / (A_iF_{ij})] \text{ (space resistance)}$$



Two-surface enclosure network (3 resistances in series)

$$Q_{12} = \sigma\,(T_1^4 - T_2^4) / [(1-\varepsilon_1)/(\varepsilon_1A_1) + 1/(A_1F_{12}) + (1-\varepsilon_2)/(\varepsilon_2A_2)]$$ 2-surface enclosure

| Configuration | $Q_{12}$ (or $Q/A$) |
|---|---|
| Infinite parallel plates | $Q/A = \sigma(T_1^4 - T_2^4) / (1/\varepsilon_1 + 1/\varepsilon_2 - 1)$ |
| Long concentric cylinders / spheres (1 inside) | $Q = \sigma A_1(T_1^4 - T_2^4) / [1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)]$ |
| Small body in large enclosure ($A_1/A_2 \to 0$) | $Q = \varepsilon_1\,\sigma\,A_1\,(T_1^4 - T_2^4)$ |
| Black surfaces ($\varepsilon = 1$) | $Q = \sigma\,A_1F_{12}(T_1^4 - T_2^4)$ |

→ 3-surface enclosures: write node equations for $J_1$, $J_2$, $J_3$; solve. A re-radiating (adiabatic) surface has $Q = 0 \to J = E_b$ (floating potential) – like a refractory furnace

wall.

## Solved Example 3 – Concentric spheres

Sphere $D_1 = 0.2$ m, $T_1 = 500$ K, $\varepsilon_1 = 0.6$, inside a sphere $D_2 = 0.4$ m, $T_2 = 300$ K, $\varepsilon_2 = 0.8$. $A_1 = 0.1257$ m$^2$, $A_1/A_2 = 0.25$.

$Q = \sigma A_1(T_1^4 - T_2^4)/[1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)] = 5.67\times10^{-8}(0.1257)(6.25\times10^{10} - 8.1\times10^{9})/[1.667 + 0.25(0.25)]$

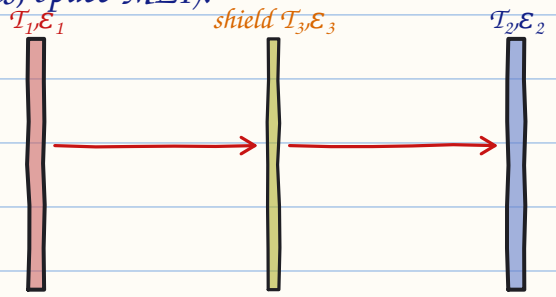
$= 387.6/1.729 =$ **224 W**

## Solved Example 4 – Parallel plates

Two large parallel plates $\varepsilon_1 = \varepsilon_2 = 0.8$, $T_1 = 800$ K, $T_2 = 400$ K: $Q/A = 5.67\times10^{-8}(800^4 - 400^4)/(1/0.8 + 1/0.8 - 1) = 21\,773/1.5 =$ **14 515 W/m$^2$**.

# Radiation Shields

A **radiation shield** is a thin, low-emissivity sheet placed between two surfaces to **increase the radiation resistance** and cut the heat flow (cryogenic tanks, thermos, thermocouple shields, space MLI).




Resistance network :

$R = (1/\varepsilon_1 - 1) + 1 + (1/\varepsilon_3 - 1) + (1/\varepsilon_3 - 1) + 1 + (1/\varepsilon_2 - 1)$

$= (1/\varepsilon_1 + 1/\varepsilon_2 - 1) + (2/\varepsilon_3 - 1)$

$$Q/A = \sigma\,(T_1^4 - T_2^4)\,/\,[\,(1/\varepsilon_1 + 1/\varepsilon_2 - 1) + \Sigma\,(2/\varepsilon_s - 1)\,]$$

If all emissivities are equal ($\varepsilon$) and $N$ shields : $Q_{N\ shields} = Q_{no\ shield} / (N + 1)$

→ Shield temperature (equal emissivities, one shield): $T_3^4 = (T_1^4 + T_2^4)/2$

→ Shield material should have **very low $\varepsilon$** (polished Al, silver, gold-coated Mylar); more shields → smaller loss but diminishing returns and need vacuum between layers.

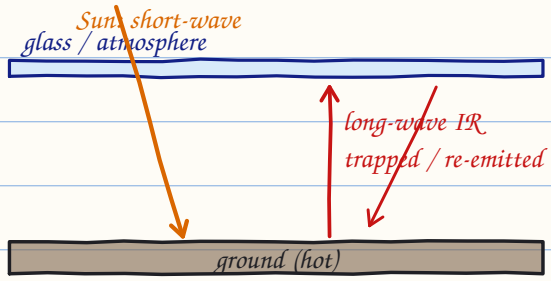
## Solved Example 5 – Effect of one shield

Plates of Example 4, add one thin shield with $\varepsilon_3 = 0.05$ on both sides: $Q/A = 21\,773/[1.5 + (2/0.05 - 1)] = 21\,773/40.5 =$ **538 W/m$^2$** – a **96 % reduction** from 14 515.

Shield temperature from $Q/A = \sigma(T_1^4 - T_3^4)/(1/0.8 + 1/0.05 - 1)$: $T_3^4 = 4.096\times10^{11} - 538(20.25)/5.67\times10^{-8} = 2.18\times10^{11}$ → $T_3 =$ **683 K**

# Greenhouse Effect

Glass, and the atmosphere, are **selectively transparent** : they transmit short-wave solar radiation ($\lambda$ < 2.5–3 $\mu$m, $\tau \approx 0.9$) but are nearly **opaque to long-wave infrared** ($\lambda$ > 3 $\mu$m) emitted by the warm ground (peak ≈ 10 $\mu$m). Energy comes in easily but cannot get out → the enclosure warms up.




Greenhouse gases : $H_2O$, $CO_2$, $CH_4$, $N_2O$, CFCs, $O_3$
Absorb at 15 $\mu$m ($CO_2$), 6–7 $\mu$m ($H_2O$) ...
Without effect : Earth ≈ -18°C
With natural effect : ≈ +15°C

**Earth energy balance** (solar constant $S$ = 1361 W/m$^2$, albedo $a$ = 0.30): absorbed $S(1-a)/4 = \sigma T_e^4$ → $238/5.67\times10^{-8} = 4.2\times10^9$ → $T_e$ = 255 K (-18°C). Actual mean surface temperature ≈ 288 K (15°C); the 33 K difference is the natural greenhouse effect.

→ **Enhanced greenhouse effect** : burning fossil fuels, deforestation, agriculture raise $CO_2$, $CH_4$, $N_2O$ → more IR absorbed → **global warming**, sea-level rise, climate change.

→ **Uses** : actual greenhouses for plants; flat-plate solar collectors (glass cover reduces re-radiation losses); car heating in sunlight.

→ **Mitigation** : renewable energy, energy efficiency, afforestation, carbon capture, reducing methane, Paris-type agreements.

## Fraction of black-body emission in a wavelength band

$$F_{0-\lambda T} = \int_0^{\lambda} E_{b\lambda}\, d\lambda \;/\; \sigma T^4 \quad \text{(function of } \lambda T \text{ only)}$$

| $\lambda T$ ($\mu$m K) | 1000 | 2000 | 2898 | 4000 | 5000 | 6000 | 8000 | 10000 |
|---|---|---|---|---|---|---|---|---|
| $F_{0-\lambda T}$ | 0.0003 | 0.067 | 0.250 | 0.481 | 0.634 | 0.738 | 0.856 | 0.914 |

**Example** : fraction of solar (5800 K) energy in visible band 0.4–0.7 $\mu$m : $\lambda T$ = 2320 → $F \approx 0.125$; $\lambda T$ = 4060 → $F \approx 0.49$. Fraction = 0.49 - 0.125 ≈ 0.37 (37 %). Remaining: UV ≈ 12 %, IR ≈ 51 %.

## Solved Example 6 – Thermocouple radiation error

A bare thermocouple ($\varepsilon = 0.8$) in a duct reads $T_t = 500$ K. Duct wall $T_w = 400$ K, $h = 50$ W/m²K. Gas temperature $T_g$?

Balance : $h(T_g - T_t) = \varepsilon\sigma(T_t^4 - T_w^4) = 0.8(5.67\times10^{-8})(6.25\times10^{10} - 2.56\times10^{10}) = 1674$ W/m²

$T_g = 500 + 1674/50 = 533$ K → radiation error = 33 K. Remedy : put a polished radiation shield around the junction, raise h (suction pyrometer).

## Radiation properties of common surfaces (approx. ε)

| Surface | ε | Surface | ε |
|---|---|---|---|
| Polished silver | 0.02 | Concrete | 0.88 |
| Polished aluminium | 0.05 | Water (thick) | 0.96 |
| Oxidised copper | 0.65 | White paint | 0.92 |
| Oxidised steel | 0.80 | Black paint / soot | 0.95–0.97 |

## Important Questions – Radiation

1. State and explain Planck's, Wien's, Stefan–Boltzmann, Kirchhoff's and Lambert's laws.
2. Define shape factor; state its properties; derive reciprocity relation.
3. Derive net radiation exchange between two gray surfaces forming an enclosure (use network).
4. What are radiation shields? Show that for N shields (equal ε) Q reduces to 1/(N+1).
5. Explain the greenhouse effect with sketch.

# Mass Transfer

Mass transfer is the movement of a chemical species from a region of higher concentration to one of lower concentration in a mixture. Examples: evaporation of water, drying of clothes, perfume spreading, absorption of gas by liquid, dissolution of sugar, humidification, membrane separation, doping of semiconductors.

→ Molecular diffusion : random molecular motion (in stagnant fluid / solids) – analogue of conduction

→ Convective mass transfer : bulk motion + diffusion between a surface and a moving

fluid – analogue of convection

| Heat transfer | Mass transfer |
|---|---|
| Driving force : temperature difference | Driving force : concentration (partial pressure) difference |
| Fourier : $q = -k\,dT/dx$ | Fick : $J = -D\,dC/dx$ |
| Thermal diffusivity $\alpha$ | Mass diffusivity $D_{AB}$ |
| $Nu = hL/k$ , $Pr = \nu/\alpha$ | $Sh = h_m L/D$ , $Sc = \nu/D$ |
| $h$ $(W/m^2K)$ | $h_m$ $(m/s)$ |

## Concentration measures

| Quantity | Definition | Unit |
|---|---|---|
| Mass concentration $\rho_A$ | mass of A / volume | $kg/m^3$ |
| Molar concentration $C_A$ | moles of A / volume $= \rho_A/M_A$ | $kmol/m^3$ |
| Mass fraction $w_A$ | $\rho_A/\rho$ | — |
| Mole fraction $x_A$ | $C_A/C$ | — |
| Partial pressure (ideal gas) | $p_A = C_A R T$ , $R = 8314$ J/kmol K | Pa |

→ Mixture : $\rho = \Sigma\rho_i$, $C = \Sigma C_i = P/RT$ ; Dalton : $P = \Sigma p_i$ ; $x_A = p_A/P$ for ideal gases.

## Fick's law of diffusion

Mass basis : $j_A = -D_{AB}\,d\rho_A/dx$ Molar basis : $J_A = -D_{AB}\,dC_A/dx$ $(kmol/m^2s)$ Fick's first law

→ $D_{AB}$ = diffusion coefficient $(m^2/s)$ of A in B. Gases ≈ $10^{-5}$, liquids ≈ $10^{-9}$, solids $10^{-10}$ – $10^{-14}$ $m^2/s$. Increases with T, decreases with P (gases).

→ Gas (Gilliland) : $D_{AB} = 4.3\times10^{-3}\,T^{3/2}\,[\,1/M_A + 1/M_B\,]^{1/2} / \{\,P\,(\,V_A^{1/3} + V_B^{1/3}\,)^2\,\}$ $cm^2/s$ (T in K, P in atm). Scaling : $D \propto T^{3/2}/P$.

→ Total molar flux in a bulk-moving mixture = diffusion + bulk flow : $N_A = -C\,D_{AB}\,dx_A/dx + x_A(N_A + N_B)$

## Conservation laws & general diffusion equation

Species conservation on a control volume (rate of accumulation = net inflow + generation) gives, with constant D, C and no reaction, Fick's second law (identical to the heat conduction eqn with $\alpha \leftrightarrow D$):

$$\partial C_A/\partial t = D_{AB}\,\nabla^2 C_A \text{ (unsteady)}$$

$$\nabla^2 C_A = 0 \rightarrow d^2C_A/dx^2 = 0 \text{ (steady, 1-D, no reaction)}$$

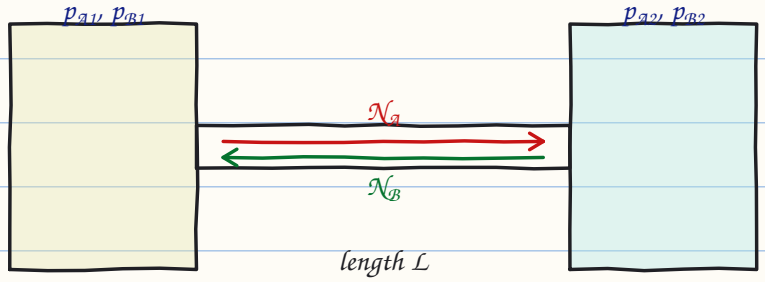
With homogeneous reaction (rate $k_1 C_A$) : $\partial C_A/\partial t = D_{AB}\nabla^2 C_A - k_1 C_A$

Overall continuity : $\partial\rho/\partial t + \nabla\cdot(\rho v) = 0$. Constitutive equations : Fick (mass), Fourier (heat), Newton (momentum) – all of the form flux = - (diffusivity) × gradient.

# Steady Molecular Diffusion in Gases

## (a) Isothermal equimolal counter-diffusion ($N_A = -N_B$)

Two large vessels at same $T$ and $P$, joined by a tube of length $L$, contain mixtures A+B in different proportions. For each mole of A moving right, one mole of B moves left, so there is *no bulk flow*:




$P = p_A + p_B = const$

Linear partial-pressure profile along tube

$J_A = -J_B$

$D_{AB} = D_{BA}$

$$N_A = D_{AB}(C_{A1} - C_{A2})/L = D_{AB}(p_{A1} - p_{A2})/(RTL)\ [kmol/m^2 s]$$ equimolal

→ Since $N_A$ is constant, $d^2C_A/dx^2 = 0$ → linear profile $C_A(x) = C_{A1} - (C_{A1} - C_{A2})\,x/L$. Mass flux = $N_A M_A$.

→ Occurs e.g. in a distillation column (binary), and in the case of "isothermal equi-mass" diffusion where molar masses are equal ($M_A = M_B$) so net mass flux is also zero.

## (b) Diffusion of A through stagnant (non-diffusing) B – Stefan's law

Evaporation of liquid A at the bottom of a narrow tube into a gas stream: B is insoluble, so $N_B = 0$ and a *bulk flow* of A carries the diffusion: $N_A = J_A + x_A N_A \rightarrow N_A = -[C D_{AB}/(1 - x_A)]\, dx_A/dz$. Integrating from $z_1$ to $z_2$ :

$$N_A = [D_{AB}P/(RTz)]\ln[(P - p_{A2})/(P - p_{A1})] = [D_{AB}P/(RTz\,p_{B,lm})](p_{A1} - p_{A2})$$

$$p_{B,lm} = (p_{B2} - p_{B1})/\ln(p_{B2}/p_{B1})$$ Stefan

→ The factor $P/p_{B,lm} \geq 1$ : diffusion through stagnant gas is *faster* than equimolal diffusion at same $\Delta p$ because of bulk (Stefan) flow. For dilute A, $p_{B,lm} \approx P$ and the two agree.

## Evaporation from a tube (liquid level falls)

Level at $z$ (from the top). Moles evaporated : $\mathcal{N}_A = (\rho_L/M_A)\, dz/dt$. With $\mathcal{N}_A$ above and $z\, dz = [\, \mathcal{D}\, \mathcal{P}\, M_A(p_{A1} - p_{A2}) / (\rho_L\, \mathcal{R}\, \mathcal{T}\, p_{B,lm})\,]\, dt$, integrate $z_0 \rightarrow z$ :

$$t = \rho_L\, \mathcal{R}\, \mathcal{T}\, p_{B,lm}\, (z^2 - z_0^2) / [\, 2\, \mathcal{D}_{AB}\, \mathcal{P}\, M_A\, (p_{A1} - p_{A2})\,]_{Winkelmann}$$

## Solved Example 6 – Equimolal counter-diffusion

Tube 0.3 m long joins two tanks at 298 K, 1 atm. $\mathcal{D}_{AB} = 2\times10^{-5}\ m^2/s$; $p_{A1} = 60.8$ kPa, $p_{A2} = 20.3$ kPa.

$\mathcal{N}_A = \mathcal{D}(p_{A1} - p_{A2})/(\mathcal{R}\, \mathcal{T}\, \mathcal{L}) = 2\times10^{-5}(40\ 530)/(8314 \times 298 \times 0.3) = 0.8106/743\ 272 = 1.09\times10^{-6}\ kmol/m^2s$

## Solved Example 7 – Evaporation of water in a tube

Water at 25°C evaporates into dry air at 1 atm from the bottom of a tube, $z = 0.15$ m to the open top. $p_{A1} = 3.17$ kPa, $p_{A2} = 0$, $\mathcal{D} = 2.6\times10^{-5}\ m^2/s$.

$p_{B1} = 98.16$ kPa, $p_{B2} = 101.325$ kPa; $p_{B,lm} = 3.17/\ln(101.325/98.16) = 99.7$ kPa

$\mathcal{N}_A = \mathcal{D}\, \mathcal{P}(p_{A1}-p_{A2})/(\mathcal{R}\mathcal{T}z\, p_{B,lm}) = 2.6\times10^{-5}(101.325)(3.17)/(8.314 \times 298 \times 0.15 \times 99.7)$

$= 2.25\times10^{-7}\ kmol/m^2s = 4.05\times10^{-6}\ kg/m^2s$

## Diffusion in liquids and solids

→ *Liquids* : $\mathcal{D}_{AB} \approx 10^{-9}\ m^2/s$ (Wilke–Chang correlation : $\mathcal{D} = 7.4\times10^{-8}(\varphi M_B)^{1/2}T/(\mu\, V_A^{0.6})\ cm^2/s$). Concentration-dependent.

→ *Solids* : interstitial/vacancy mechanism; $\mathcal{D} = \mathcal{D}_0 \exp(-Q/\mathcal{R}T)$ (Arrhenius) e.g. carbon in steel, hydrogen in metals, drying of wood.

→ Steady diffusion through a plane membrane/wall (thickness $L$): $\mathcal{N}_A = \mathcal{D}_{AB}(C_{A1} - C_{A2})/L = \mathcal{D}\, S\, (p_{A1} - p_{A2})/L$ ($S$ = solubility, Henry's law). Through a cylinder : $\mathcal{N}_A$ per unit length $= 2\pi\mathcal{D}(C_{A1}-C_{A2})/\ln(r_2/r_1)$.

# Convective Mass Transfer

Mass transfer between a surface and a moving fluid is described by a *mass-transfer coefficient* $h_m$ (analogous to $h$):

$$\mathcal{N}_A = h_m\, (C_{A,s} - C_{A,\infty})\ [kmol/m^2s]\ \ h_m \text{ in } m/s$$

$$\text{Mass flow } \dot{m}_A = h_m\, \mathcal{A}\, (\rho_{A,s} - \rho_{A,\infty})$$

| Number | Definition | Meaning | Heat analogue |
|---|---|---|---|
| Sherwood Sh | $h_m L/D_{AB}$ | convective / diffusive mass transfer | Nu |
| Schmidt Sc | $\nu/D_{AB}$ | momentum / mass diffusivity | Pr |
| Lewis Le | $\alpha/D_{AB} = Sc/Pr$ | thermal / mass diffusivity | — |
| Stanton (mass) $St_m$ | $h_m/V = Sh/(Re\ Sc)$ | | St |

Correlations are obtained from the heat transfer ones by replacing $Nu \to Sh$ and $Pr \to Sc$:

| Case | Correlation |
|---|---|
| Laminar flat plate (local) | $Sh_x = 0.332\ Re_x^{1/2}\ Sc^{1/3}$ |
| Laminar flat plate (average) | $Sh_L = 0.664\ Re_L^{1/2}\ Sc^{1/3}$ |
| Turbulent flat plate | $Sh_L = 0.037\ Re_L^{0.8}\ Sc^{1/3}$ |
| Turbulent flow in a tube | $Sh_D = 0.023\ Re_D^{0.83}\ Sc^{1/3}$ (Gilliland–Sherwood : $Sc^{0.44}$) |
| Sphere (Frössling) | $Sh_D = 2 + 0.6\ Re_D^{1/2}\ Sc^{1/3}$ |
| Free convection (vertical plate) | $Sh = C\ (Gr_m\ Sc)^n$ |

→ Chilton–Colburn analogy : $St\ Pr^{2/3} = St_m\ Sc^{2/3} = C_f/2 \to h/h_m = \rho\ c_p\ Le^{2/3}$ ($h_m$ from known $h$, e.g. wet-bulb thermometer, cooling towers)

→ Boundary layers: $\delta/\delta_c = Sc^{1/3}$ – the concentration BL thickness plays the role of $\delta_t$.

## Solved Example 8 – Evaporation from a water surface

Air at 25°C, 2 m/s flows over a 1 m long water surface. $\nu = 1.55\times10^{-5}$, $D = 2.6\times10^{-5} \to Sc = 0.596$, $Re_L = 1.29\times10^5$ (laminar).

$Sh_L = 0.664(359)(0.596)^{1/3} = 200.7$; $h_m = Sh\ D/L = 200.7(2.6\times10^{-5})/1 = 5.2\times10^{-3}$ m/s

$C_{A,s} = p_v/RT = 3.17/(8.314 \times 298) = 1.28\times10^{-3}$ kmol/m³ ; dry air $C_{A,\infty} = 0$ : $N_A = 5.2\times10^{-3}(1.28\times10^{-3}) = 6.7\times10^{-6}$ kmol/m²s ≈ 0.43 kg/m²h

## Solved Example 8B – Heat–mass analogy

Air at 25°C over a surface gives $h = 20$ W/m²K. $Sc = 0.596$, $Pr = 0.70 \to Le = 0.851$; $Le^{2/3} = 0.898$. From the analogy : $h_m = h/(\rho\ c_p\ Le^{2/3}) = 20/(1.2 \times 1007 \times 0.898) =$ 0.0184 m/s.

## Solved Example 9 – Diffusion through a solid wall

Hydrogen diffuses through a steel plate 5 mm thick; $C_{A1} = 2.0$ and $C_{A2} = 0.5$ kg/m$^3$; $D = 2.6\times10^{-13}$ m$^2$/s. $N_A = D(C_{A1} - C_{A2})/L = 2.6\times10^{-13}(1.5)/0.005 = 7.8\times10^{-11}$ kg/m$^2$s (very slow in solids).

## Important Questions – Mass Transfer

1. State Fick's law of diffusion. Compare heat and mass transfer analogy.
2. Derive an expression for molar flux in equimolal counter diffusion.
3. Derive Stefan's expression for diffusion of A through stagnant B. Where is it used?
4. Define Sherwood, Schmidt and Lewis numbers; give a correlation for $h_m$.

## Quick Revision

| Item | Formula |
|---|---|
| Black-body emissive power | $E_b = \sigma T^4$ |
| Wien | $\lambda_{max} T = 2898\ \mu m\cdot K$ |
| View factor | $A_1F_{12} = A_2F_{21}$ ; $\Sigma F_{1j} = 1$ |
| Two-gray-surface exchange | $\sigma\Delta(T^4)$ / [$\Sigma$ resistances] |
| Shields (equal $\varepsilon$) | $Q = Q_0/(N+1)$ |
| Fick | $N_A = -D\ dC/dx$ |
| Equimolal | $N_A = D(p_{A1}-p_{A2})/RTL$ |
| Stagnant B | extra factor $P/p_{B,lm}$ |
| Sh, Sc | $h_mL/D$ , $\nu/D$ |

